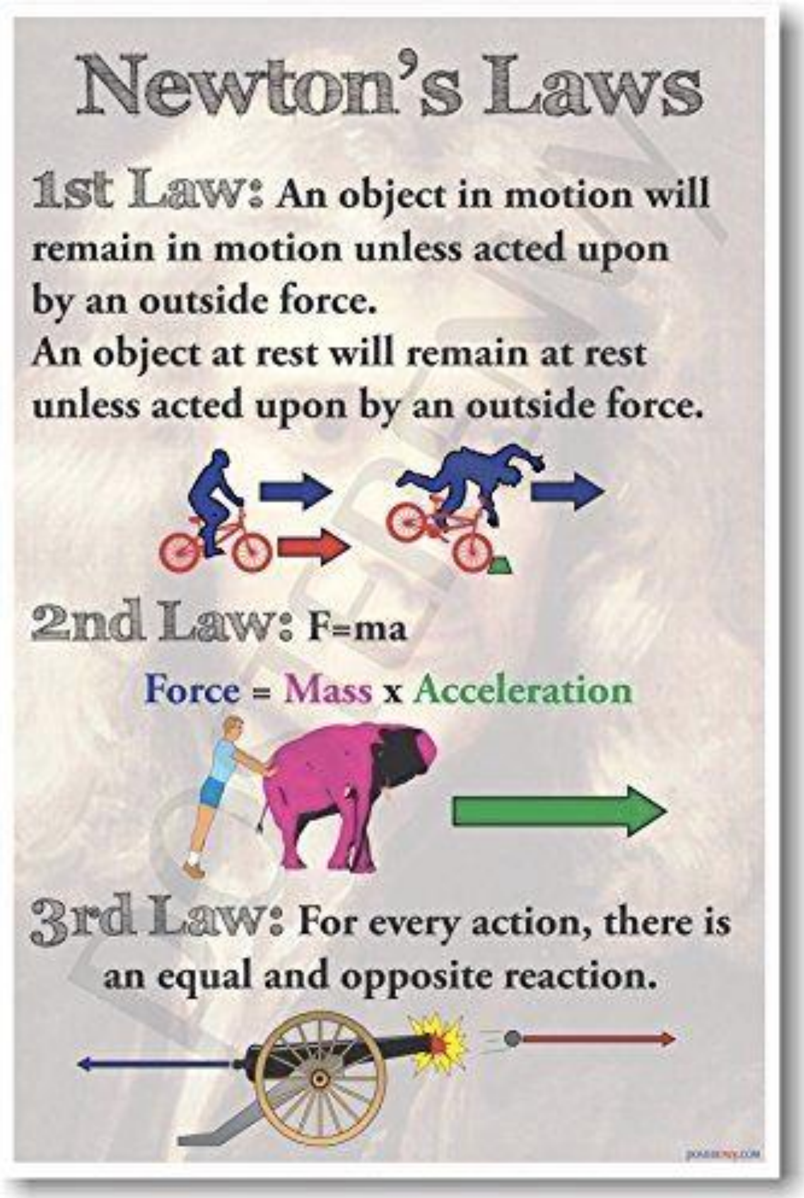
YEAR 10 SCIENCE

PHYSICAL SCIENCES
(Extension)

FORCES AND MOTION

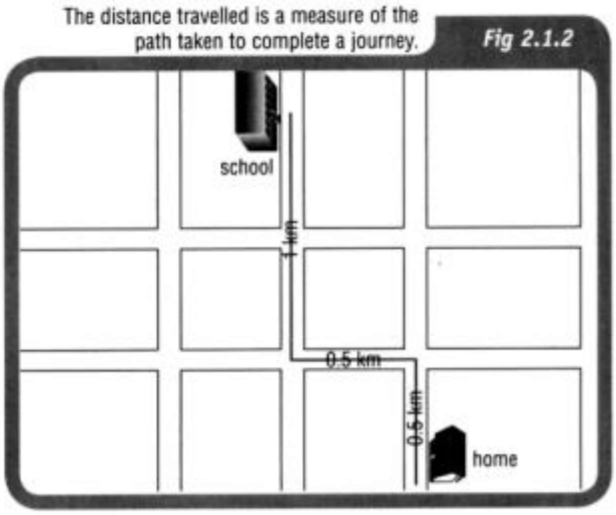
NAME: ______________________________



DISTANCE AND DISPLACEMENT

Distance (d): how far a body moves in any direction.

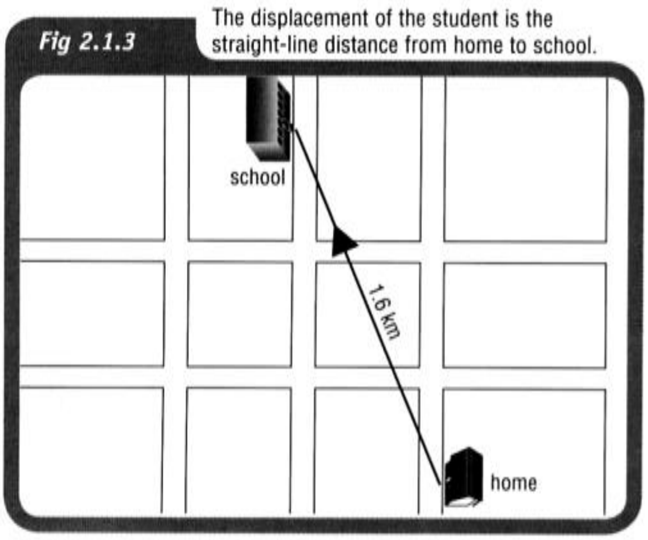
e.g. The diagram below shows the movement of a student between home and school.



The distance travelled is 2.0 km.

Displacement (s): change of position of a body from the start of the motion to the finish of the motion, and is in a particular direction.

e.g. The diagram below shows the displacement of the student when travelling to school.



The straight-line distance is 1.6 km. No direction is indicated in the diagram.

If the distance you walk from home to school is 2.0 kilometres, the distance you travel from home to school and back home again is 4.0 kilometres. If you start at home and finish the day's journey at home, your starting and finishing points are the same position (home) then your ***displacement for the day is actually 0 km***.

In Physics, distance is known as a ***scalar quantity*** and displacement is a ***vector quantity***.

Vectors have a ***direction*** associated with them.

Worksheet - Distance Versus Displacement

1. Map 171 - use the scale to work out the **distance and displacement** from HCC to the roundabout at The Boulevarde and Pinaster Parade. Check your measurements using Google maps.

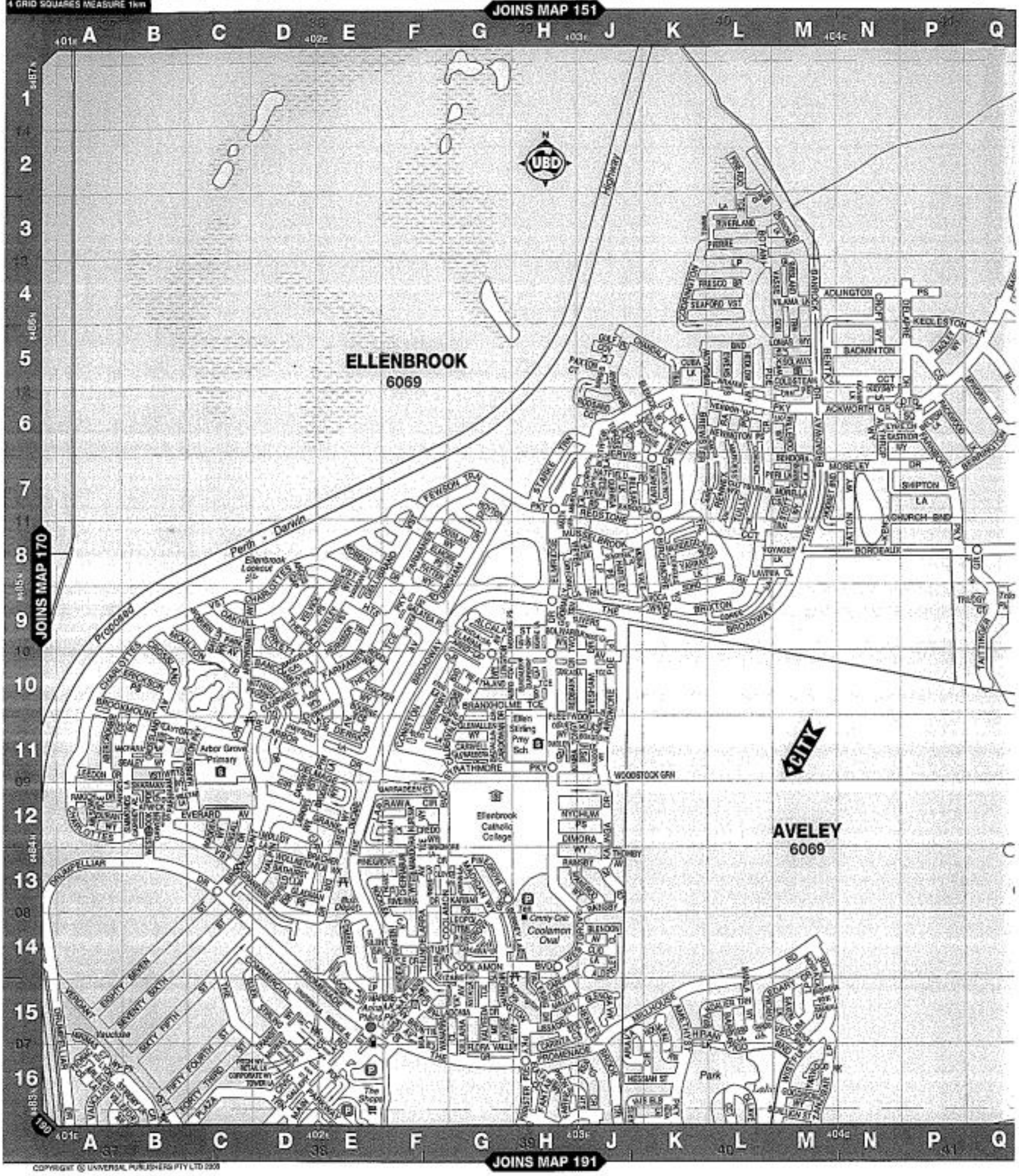
   Working

2. Map 185 - use the scale to work out the **distance and displacement** from Whitfords City shops (Endeavour Road entrance) to the roundabout at Whitfords Avenue and Angove Drive. Check your measurements using Google maps.

   Working

MAP **171**

4 GRID SQUARES MEASURE 1km



SCALE 1:19 000 — 1km
Metres 500 1000

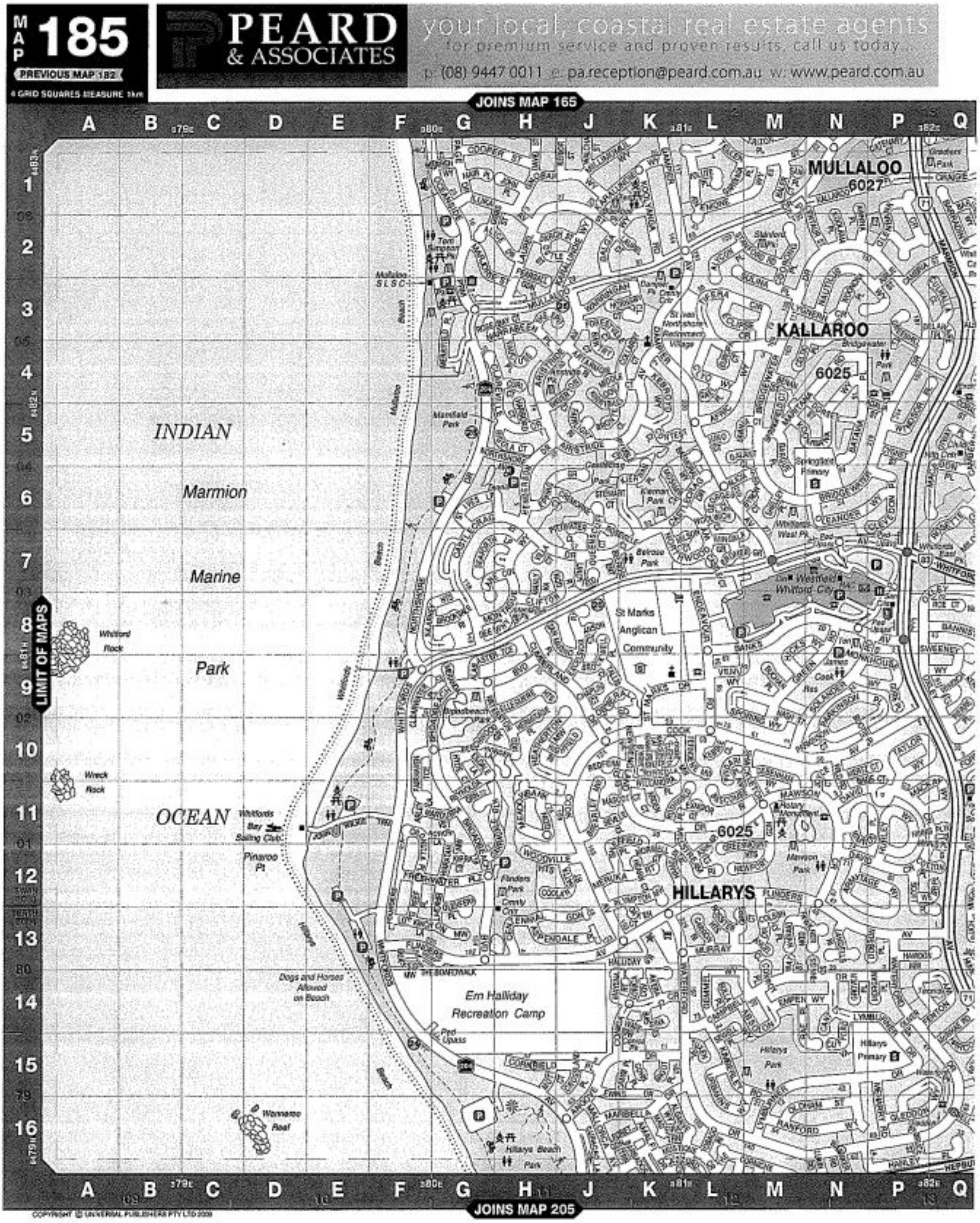
FREEWAY
PROPOSED FREEWAY
HIGHWAY or MAIN ROUTE
ALTERNATE ROUTE
TRAFFICABLE ROAD
PROPOSED ROAD

PARK, RESERVE, OVAL
SCHOOL, HOSPITAL
MISCELLANEOUS AREA
MALL, PLAZA
SWAMP

AMBULANCE STATION
BARBECUE
BOAT RAMP
BOWLING CLUB/GREEN
CAMPING AREA
CARAVAN PARK

CAR PARK
CYCLEWAY
DISTANCE FROM GPO
EMERGENCY TELEPHONE
EXPRESS POST BOX
FIRE STATION F

PRIMARY RECTANGLE SWAN BG35



SCALE 1:19 000 — 1km
Metres 500 1000

FREEWAY	PARK, RESERVE, OVAL	AMBULANCE STATION	CAR PARK
PROPOSED FREEWAY	SCHOOL, HOSPITAL	BARBECUE	CYCLEWAY
HIGHWAY or MAIN ROUTE	MISCELLANEOUS AREA	BOAT RAMP	DISTANCE FROM GPO
ALTERNATE ROUTE	MALL, PLAZA	BOWLING CLUB/GREEN	EMERGENCY TELEPHONE
TRAFFICABLE ROAD	SWAMP	CAMPING AREA	EXPRESS POST BOX
PROPOSED ROAD		CARAVAN PARK	FIRE STATION

PRIMARY RECTANGLE SWAN BG35
PRIMARY RECTANGLE PERTH BG34

[redacted]

YEAR 10 SCIENCE
PHYSICAL SCIENCES
WORKSHEET - DISTANCE AND DISPLACEMENT

Use Google maps to do the following determinations of distance and displacement.

1. (a) Find the distance from HCC to Bassendean Train Station.

   Distance = ______________________

   (b) Blow the map of the trip up as large as possible on your screen, then print it out in the LRC.

   (c) Using a ruler and protractor, and the scale on the bottom right of the map, to determine the displacement. Use the space below for your working.

   Remember to work out the direction. e.g. W 35° N or bearing 305°.

   Displacement = ____________________________________

2. Repeat the steps for HCC to the Benedictine Monastery in New Norcia.

   Distance = ______________________

   Displacement = ____________________________________

3. Determine the distance and direction from Perth Airport to Darwin Airport. Use the ***car symbol***, indicating that you are driving between the two.

Distance = ____________________

Displacement = _________________________________

VECTORS

SCALAR QUANTITIES - have magnitude or size only.

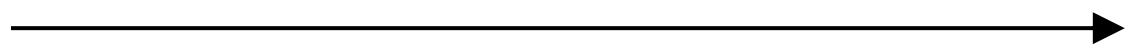
e.g. mass, distance, speed, density, temperature.

VECTOR QUANTITIES - have magnitude and direction.

e.g. displacement, velocity, acceleration, force, impulse, momentum.

Vectors are represented by arrows - the length indicates the magnitude of the vector and the arrowhead indicates the direction in which it acts.

i.e.



This is its direction.

The length gives the magnitude.

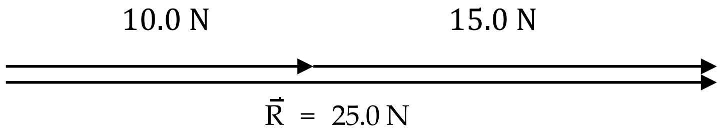
COMBINING VECTORS

ADDITION

> Vectors are added head-to-tail. The resultant vector runs from the tail of the first vector to the head of the last vector.

1. **ADDITION**

ONE-DIMENSION

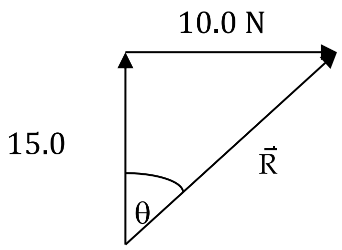


The resultant vector R = 25 .0 N - it runs from the tail of the first vector to the head of the last vector.

TWO DIMENSIONS

VECTORS AT RIGHT ANGLES

Use Pythagorus' Theorem and trigonometry to find the resultant.



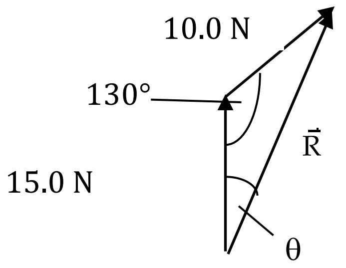
$$\vec{R} = \sqrt{(10.0)^2 + (15.0)^2} = 18.03\text{ N}$$

$$\tan\theta = \frac{10.0}{15.0} \quad \Rightarrow \quad \theta = 33.69^\circ$$

$\therefore$ <u>$\vec{R}$ = 18.0 N at 33.7° to the 15.0 N vector</u>.

VECTORS NOT AT RIGHT ANGLES

Use the sine and cosine rules for triangles to find the resultant vector.



$$\vec{R} = \sqrt{(15.0)^2 + (10.0)^2 - 2(15.0)(10.0)\cos 130°}$$

$$= 22.76 \text{ N}$$

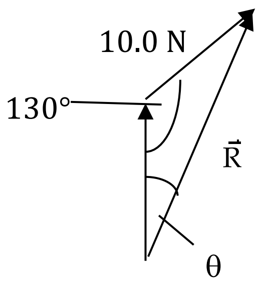
$$\frac{10.0}{\sin \theta} = \frac{22.76}{\sin 130°}$$

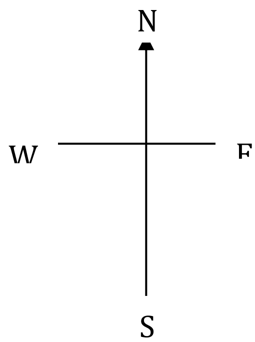
$$\Rightarrow \theta = 19.67°$$

$\therefore$ <u>$\vec{R}$ = 22.8 N at 19.7° to the 15.0 N vector</u>.

NOTE
If compass directions are given, it makes it easier to describe the direction.

Consider the example above.





The resultant vector R is at a 19.7° angle east of north.

∴ Direction is N 19.7° E.

Also, north can be considered the 0° bearing, with all other bearings determined clockwise from this direction.

∴ Direction is a bearing of 19.7°.

<u>Hence R = 22.8 N N 19.7° E or R = 22.8 N on a bearing of 19.7°</u>.

Year 10 Science
Physical Sciences
Worksheet - Displacement (Extension)

RESULTANT OF TWO (OR MORE) DISPLACEMENTS

When a body is subjected to two (or more) displacements the final direct distance (the displacement) from the starting position is *NOT* equal to the total distance travelled (unless the individual displacements are in the same direction).

To determine the exact location (distance and direction from the starting position) of a body, it is necessary to determine the *RESULTANT* displacement.

A resultant of two (or more) vectors may be defined as the single vector which on its own will produce the same effect as the two (or more) vectors combined.

DETERMINATION OF RESULTANTS

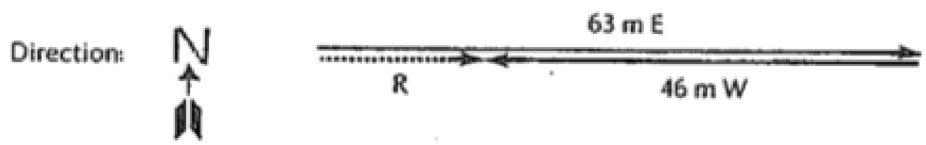
To determine the resultant:

1. Place the vectors head to tail (in any order).
2. Draw in the resultant which is from the tail of the first vector to the head of the last.
3. If the vectors have been drawn to scale then the magnitude of the resultant and the direction can be measured — otherwise both can be calculated.

This method is often referred to as the *"head to tail"* or *"triangle"* method.

TYPE EXAMPLE 5: What is the resultant displacement of a displacement of 63 m east and a displacement of 46 m west?

Sketch vectors head to tail (approximately to scale) and the resultant displacement.



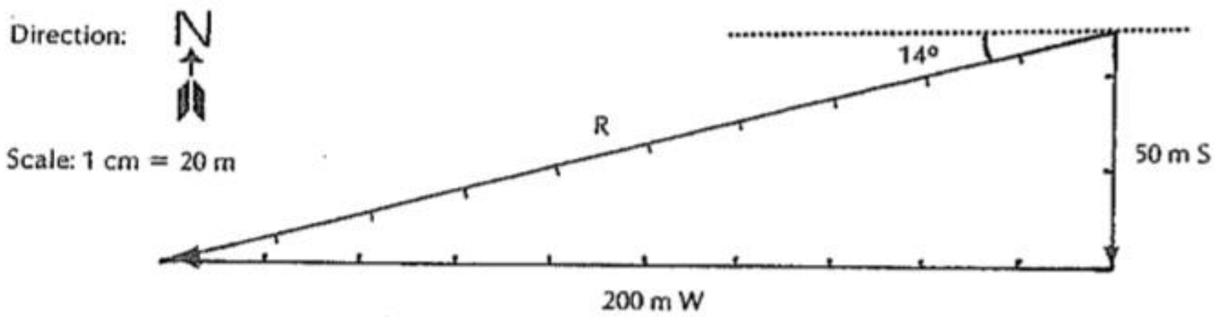
For displacements in one straight line, measurement (i.e. graphical solution) is unnecessary as the calculation of the magnitude of the resultant is obvious.

$$\text{resultant, } R = (63 \text{ m}) - (46 \text{ m}) = 17 \text{ m}$$

i.e. the resultant displacement is 17 m east.

TYPE EXAMPLE 6: Determine the resultant displacement if a person walks 50 m south to a corner and then 200 m west to go to the local store.

A. GRAPHICAL SOLUTION

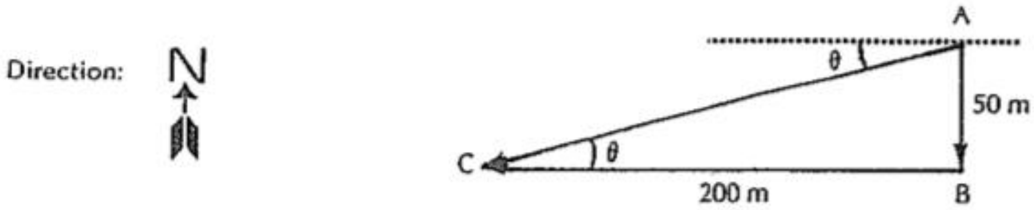


$$\text{length of R} = 10.3 \text{ cm}$$
$$\therefore \text{magnitude of R} = (10.3 \times 20) \text{ m} = 206 \text{ m}$$

i.e. the resultant displacement is 206 m W 14° S

B. CALCULATION SOLUTION (right angled triangles only).

1. Sketch the vector triangle (approximately to scale).



2. Using Pythagorus' Theorem calculate a value for the magnitude of the resultant, AC.

AC = resultant
= ?
AB = 50 m S
BC = 200 m W

$$AC^2 = AB^2 + BC^2$$
$$= (50)^2 + (200)^2$$
$$= 2\,500 + 40\,000$$
$$= 42\,500$$
$$\therefore AC = 206.2 \text{ m}$$

3. Using trigonometric ratios determine the direction of the resultant.

$$\tan\theta = \frac{AB}{BC} = \frac{50}{200} = 0.25$$
$$\therefore \theta = 14°$$

i.e. the resultant displacement is 206.2 m W 14° S.

SET 5: RESULTANT DISPLACEMENT III

Using Pythagorus' Theorem and trigonometric ratios determine the resultant displacement for each of the following:

1. 300 km N, 300 km W.
2. A climb 6 m out of a well and 8 m away from it.
3. 500 m E, 1 200 m S.
4. 17.32 m horizontally then 10 m vertically upwards.
5. 1.2 km NW, 1.6 km NE.

SET 5

1. 424 km NW **2.** 10 m 53° to the vertical **3.** 1 300 m E 67° S
4. 20 m 30° to the horizontal **5.** 2 km N 8° E

SPEED AND VELOCITY

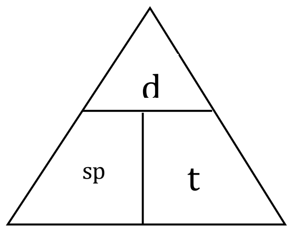
Speed: distance moved per unit time.

$$\boxed{\text{Speed} = \frac{d}{t}}$$

where speed = how fast the object travels for the entire journey (m/s)
d = distance travelled for the entire journey (m)
t = time taken for the entire journey (s)

Speed is a ***scalar*** quantity as direction is ***not*** important. Velocity is a ***vector*** quantity and has a ***direction*** involved.

Use the triangle shown to help calculate speed.

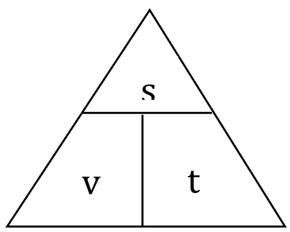


Velocity: displacement moved in a given direction per unit time.

$$\boxed{v = \frac{s}{t}}$$

where v = velocity in a given direction (m/s)
s = displacement in a given direction (m)
t = time taken to move in a given direction (s)

Use the triangle shown to help calculate velocity.



Note

The units for speed and velocity are metres per second (m/s), not km/h.

To convert km/h to m/s, divide the reading in km/h by 3.6.

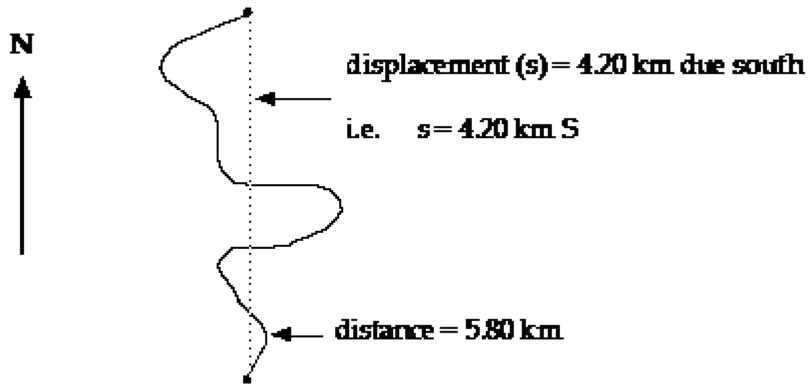
Example 1

$$60.0 \text{ km/h} = \frac{60.0}{3.6}$$
$$= 16.7 \text{ m/s}$$

Example 2

A cyclist rides along a coast road that meanders and changes direction constantly. He covers a distance of 5.80 km in a time of 12.2 minutes. However the straight line distance between his start and finish points is only 4.20 km in a direction due south. Calculate:

(a) the average speed of the cyclist.
(b) his average velocity.



(a)

$$\text{Speed} = \frac{d}{t}$$
$$= \frac{5800}{732}$$
$$= 7.923 \text{ m/s}$$

\ speed = 7.92 m/s

(b)

$$v = \frac{s}{t}$$
$$= \frac{4200}{732}$$
$$= 5.738 \text{ m/s}$$

\ v = 5.74 m/s South

The direction is important - make sure you put it in!

Speed and velocity

Student: .... Class: ....

Calculating average speed

1. The average speed (v) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

where d = distance travelled and t = time taken.

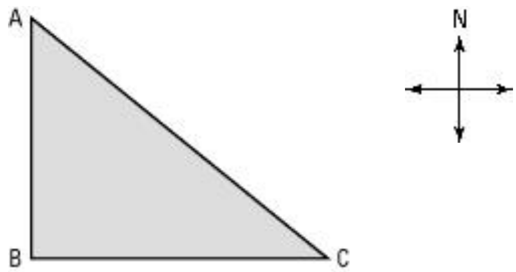
Complete the following table. Make sure you include the appropriate units for each quantity.

Distance travelled	Time taken	Average speed
100 m	20 s	
3 km		60 km/h
	15 s	20 m/s
2 km		25 m/s

Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.

AB = 600 m
BC = 800 m



Both vehicles take 70 s for the trip.
Perform the necessary calculations to complete the following table.

Average speed of X	Average velocity of X	Average speed of Y	Average velocity of Y

Year 10 Science
Physical Sciences
Worksheet - Speed and Velocity

SPEED

The speed of a body is the rate of change of distance by the body.

The average speed of a body over a distance travelled in a time interval can be determined by:

$$\boxed{\text{Speed (av)} = \frac{d}{t}}$$

where d = distance (measured from zero)
t = time interval (measured from zero)

Speed is a scalar quantity as it involves no specific direction. Distance is the total length of the path covered by the body.

If to accurately describe motion direction is required, then *VELOCITY* is used in place of speed.

VELOCITY

The velocity of a body is the rate of change of displacement of the body.

The average velocity of a body over a displacement in a measured time interval can be determined by:

$$\boxed{v_{av} = \frac{s}{t}}$$

v_{av} = average velocity
s = displacement (measured from zero)
t = time interval (measured from zero)

Velocity is a vector quantity. *The direction of the velocity is that of the displacement (unless otherwise stated).*

UNITS OF VELOCITY AND SPEED

Velocity and speed are both determined by dividing units of length (m) by units of time (s).

Units: $\frac{\text{metres}}{\text{seconds}} = \frac{m}{s} = m\ s^{-1}$ (or m/s)

DETERMINATION OF DISPLACEMENT

The equation for average velocity can be rearranged to determine the displacement.

$$v_{av} = \frac{s}{t}$$

rearranging gives: $s = v_{av} \times t$
similarly $d = \text{speed (av)} \times t$

TYPE EXAMPLE 7: Calculate the average velocity of a vehicle which travels 3 km east in 5 minutes.

$v_{av} = ?$
$s = 3$ km
$= (3 \times 1\,000)$ m
$t = 5$ minutes
$= (5 \times 60)$ s

$$v_{av} = \frac{s}{t} = \frac{3 \times 1\,000}{5 \times 60} = 10\ m\ s^{-1}$$

i.e. the average velocity is 10 m s^{-1} east.

TYPE EXAMPLE 8: Determine the displacement of a body which moves with a uniform velocity of 2.4 m s^{-1} for 50 s.

$v_{av} = 2.4 \text{ m s}^{-1}$
$s = ?$
$t = 50 \text{ s}$

$$v_{av} = \frac{s}{t}$$
$$\therefore \quad s = v_{av} \times t$$
$$= 2.4 \times 50$$
$$= 120 \text{ m}$$

i.e. the displacement is 120 m in the direction of the velocity.

SET 6: VELOCITY I

1. What is the average speed of a sprinter who runs 100 m in 12.5 s?
2. A car takes 80 s to complete one lap of a 2 km circuit. What is its average speed (in m s^{-1})?
3. A body travels 36 m in 4 s. What is its average speed?
4. How far will a bird fly in 25 s travelling with an average speed of 6 m s^{-1}?
5. A person swims from one end of a pool to the other at an average velocity of 1.5 m s^{-1} in 20 s. What is the length of the pool?
6. Determine the distance travelled by a plane in 1 hour if it flies due south with a uniform velocity of 90 m s^{-1}
7. How long will it take for a body moving with a uniform velocity of 7 m s^{-1} to cover 98 m?
8. A cricket ball is thrown a distance of 90 m from the boundary to the wicket at an average horizontal speed of 30 m s^{-1}. How long will it take for the ball to reach the wicket?
9. Determine the time taken for a horse running with an average speed of 14 m s^{-1} to complete one circuit of a 840 m track.
10. How long will it take to complete a trip from Perth to Kalgoorlie, a distance of 595 km, travelling at an average speed of 70 km h^{-1}?
11. A train moving with an average speed of 80 km h^{-1} takes 0.25 hours to travel from one railway siding to the next. What is the distance between the two sidings?
12. Determine the average speed (km h^{-1}) of a plane which flies a distance of 720 km in 1.5 hours?

SET 6

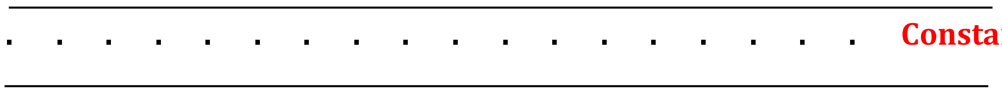
1. 8 m s^{-1}	**2.** 25 m s^{-1}	**3.** 9 m s^{-1}	**4.** 150 m	**5.** 30 m
6. 324 000 m	**7.** 14 s	**8.** 3 s	**9.** 60 s	**10.** 8.5 h
11. 20 km	**12.** 480 km h^{-1}			

TICKER TIMERS

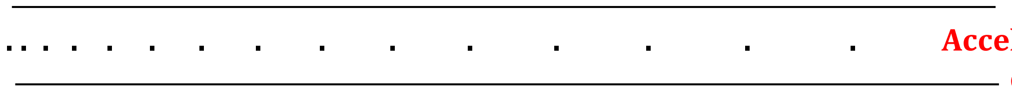
A ticker timer will mechanically record 50 dots every second.

> ∴ 1 dot spacing represents 1/50th of a second = 0.02 s

Visually, the ticker tape shows whether the motion represented is constant speed or acceleration.



Constant speed - dots evenly spaced.

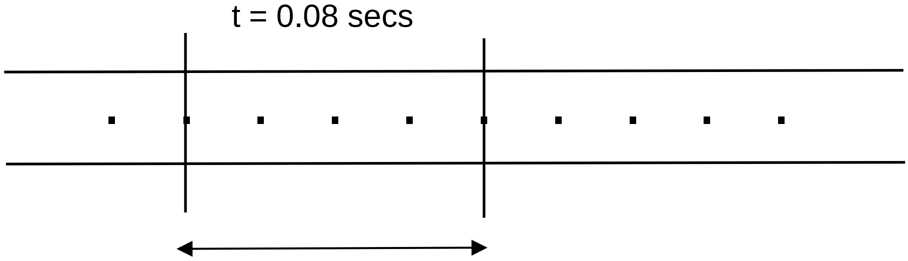


Acceleration - dots get further apart.

Calculations

Constant speed

- Choose 1 dot as your start point.
- Count 4 dots from this point- this represents 0.10 seconds.
- Measure the distance from the start to the fifth dot.
- Calculate the speed or velocity using $v = \frac{d}{t}$.



$v = ?$

$d = 11.3 \text{ cm} = 0.113 \text{ m}$

$t = 0.08 \text{ s}$

$$v = \frac{d}{t} = \frac{0.113}{0.08} = 1.41 \text{ m/s}$$

\ <u>speed (velocity) = 1.41 m/s</u>

Acceleration

- Choose 5 dots near the start of the tape - this represents 4 dot spacings = 0.08 seconds.

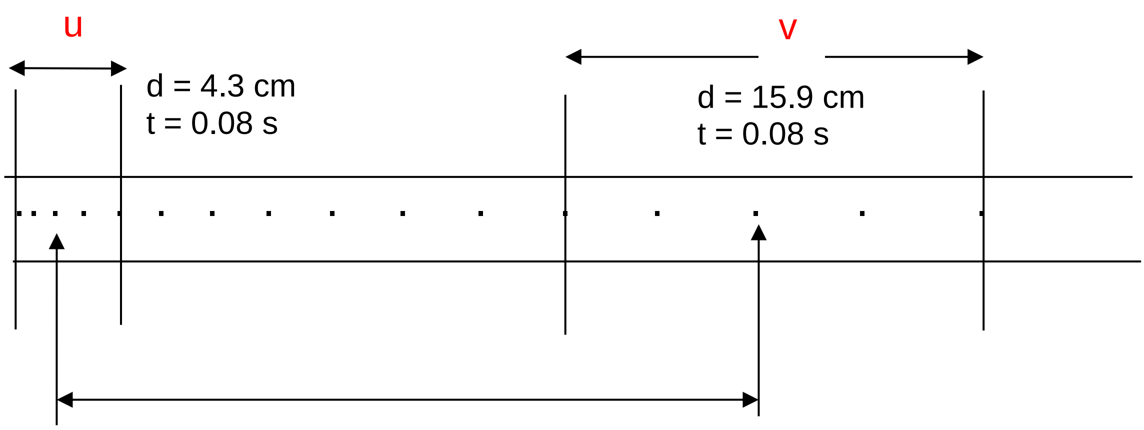
- Measure the distance for the 5 dots and calculate the average speed as above. This is the **initial velocity (u)**.

- Select 5 dots near the end of the tape. Measure the distance for these and calculate the average speed - this is the **final velocity (v)**.

- Mark the middle dot of the "u" section and middle dot of the "v" section.

  Count the number of dot spacing between them and multiply by 0.02 seconds to work out the time.

- Calculate the acceleration using $a = \dfrac{v - u}{t}$.



t = 11 spacings
= 11 x 0.02
= 0.22 secs

To calculate u:

$u = \dfrac{d}{t}$

d = 4.3 cm = 0.043 m

$= \dfrac{0.043}{0.08}$

t = 0.08 s

= 0.538 m/s

To calculate v:

$v = \dfrac{d}{t}$

d = 15.9 cm = 0.159 m

$= \dfrac{0.159}{0.08}$

t = 0.08 s

= 1.99 m/s

To calculate the acceleration:

v = 1.99 m/s

$a = \dfrac{v - u}{t}$

u = 0.538 m/s

$= \dfrac{1.99 - 0.538}{0.22}$

a = ?

$= 6.60\ m/s^2$

t = 0.22 s

\ acceleration = 6.60 m/s^2 forwards

Worksheet 8.2

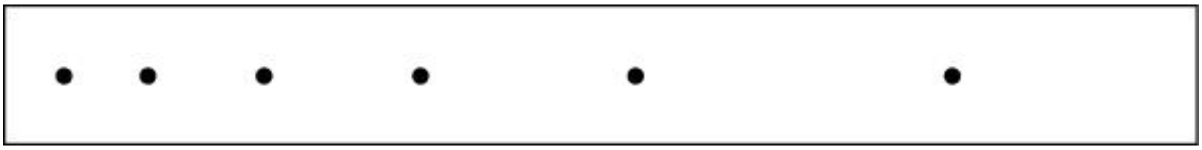
Ticker tapes

Student: .... Class: ....

Describing motion

1. For each ticker tape shown, describe the motion of the body to which the tape was attached. The left-hand dot is at time zero.

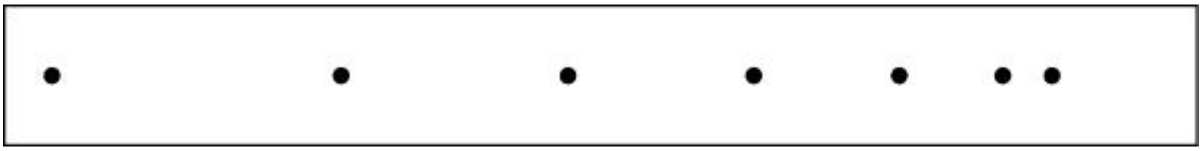
(a)



(b)

Measuring speed

2. The time between each of the dots on the following ticker tape is 0.02 s.



(a) Calculate the average speed of the object to which the tape was attached for each time interval. Express your answers in mm/s.

Interval	Distance between dots (mm)	Average speed (mm/s)
AB		
BC		
CD		
DE		
EF		
FG		

(b) Describe the motion of the object to which this tape was attached.

INVESTIGATION 8.1

Ticker timer tapes

AIM To record and analyse motion using a ticker timer

Materials:
ticker timer
power supply
scissors
G-clamp
ticker tape (in 60 cm lengths)

METHOD AND RESULTS

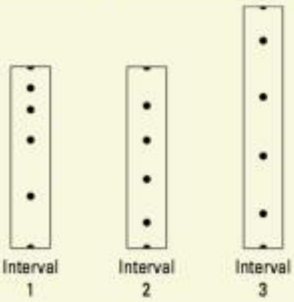
- Clamp the ticker timer firmly to the edge of a table or bench so that you will be able to pull 60 cm of ticker tape through it. Connect the ticker timer to the AC terminals of the power supply and set the voltage as instructed by your teacher.
- Thread one end of the ticker tape through the ticker timer so that it goes under the carbon paper disc.
- Turn on the power supply and check that the ticker timer leaves a black mark on the ticker tape.
- Hold the end of the ticker tape and walk away from the ticker timer so that the ticker tape moves through at a steady speed.
- Remove the ticker tape and mark off the first clear dot made and every fifth dot after the first. (There should be four dots between each of the marked-off dots on the ticker tape.)

1. Measure the distance travelled during each 0.1 s interval and write it on your tape. Label the intervals as interval 1, interval 2, interval 3, etc.

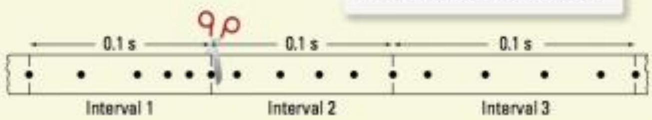
- Cut your ticker tape into 0.1 s intervals and glue the strips in order onto a sheet of paper. Each strip shows the distance travelled during a 0.1 s time interval. The graph therefore shows how the speed changed with time.

DISCUSS AND EXPLAIN

2. How much time elapsed between the printing of the first clear dot and the last dot marked off?
3. Calculate the average speed for the motion that took place between the printing of the first clear dot and the last marked dot.
4. Calculate the average speed during each 0.1 s interval.
5. Did you succeed in keeping your speed steady?



Using ticker tape to plot a graph



UNDERSTANDING AND INQUIRING

REMEMBER

1 Explain the difference between instantaneous speed and average speed. Use an example to support your explanation.

2 Which methods are used by the police to measure:
(a) average speed
(b) instantaneous speed?

THINK

3 Make a list of reasons why a speedometer reading might not be accurate. Include anything that could change the diameter of the vehicle's tyres.

4 After being phased out, digitectors are making a comeback in police detection of speeding drivers. Suggest advantages they might have over radar and laser guns.

PROCESS AND ANALYSE

5 Calculate the average speed during the second and third 0.1 s intervals of the ticker tape shown in Investigation 8.1.

INVESTIGATE

6 Use data-logging equipment with a motion detector or light gates to record the motion of a toy car or cart down a slope. Use the software to produce a graph of distance versus time and a graph of speed versus time. Comment on the shape of your graphs.

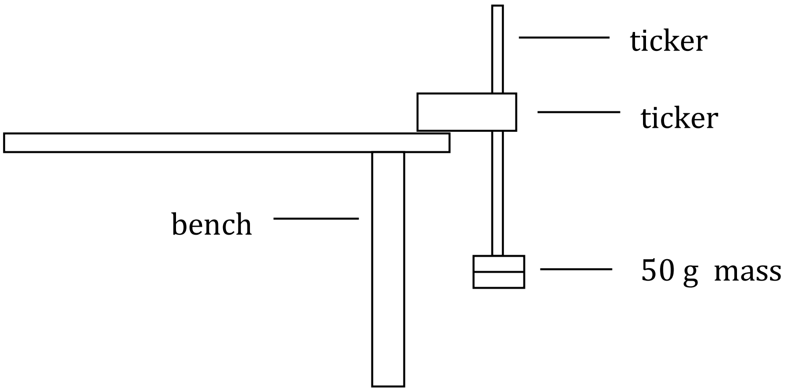
eBook plus — work sheets: 8.1 Speed and velocity; 8.2 Ticker tapes

YEAR 10 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION: ACCELERATION DUE TO GRAVITY

AIM To determine the acceleration due to gravity of an object..

METHOD

1. Set up the following apparatus.



2. Start the timer and drop the mass through the timer.
3. Analyse and record the tape to determine the acceleration.
4. Repeat several trials and average your results.

RESULTS Record your results appropriately. Show your working.

DISCUSSION Analyse the activity and list possible sources of error in your recordings.

Suggest an alternative method that may give a more precise measurement.

CONCLUSION This must answer the aim.

ACCELERATION

Acceleration (a)*:* rate of change of velocity in a given direction.

It is a vector quantity and is calculated by the equation:

$$\boxed{a = \frac{v - u}{t}}$$

where
- a = acceleration in a given direction (m/s^2)
- u = initial velocity of the object (m/s)
- v = final velocity of the object (m/s)
- t = time taken for the change in velocity to occur (s)

This equation can also be written as:

$$\boxed{v = u + at}$$

Note

- When solving problems, ***pick one direction as the positive direction***.
- Any vector in this direction is a positive value.
- Any vector in the ***opposite direction*** is a ***negative value***.

Examples

1. A car moving at 30.0 km/h along a straight road accelerates uniformly to 70.0 km/h in 12.0 s. Calculate the acceleration experienced.

v = 70.0 km/h
= 19.44 m/s

u = 30.0 km/h
= 8.333 m/s

a = ?

t = 12.0 s

Take forwards as positive.

$$a = \frac{v - u}{t} = \frac{19.44 - 8.333}{12.0} = 0.9256\ m/s^2$$

$\therefore$ <u>$a = 0.926\ m/s^2$ forwards</u>

2. A car moving at 70.0 km/h along a straight road brakes suddenly to avoid a dog that ran onto the road. It reduced its velocity to 15.0 km/h in 2.97 s and successfully missed the animal. Calculate the deceleration of the car.

$v = 15.0$ km/h

$\quad = 4.167$ m/s

$u = 70.0$ km/h

$\quad = 19.44$ m/s

$a = ?$

$t = 12.0$ s

Take forward as positive.

$$a = \frac{v - u}{t}$$

$$= \frac{4.167 - 19.44}{2.97}$$

$$= -\ 5.142 \text{ m/s}^2$$

\ $\underline{a = 5.142 \text{ m/s}^2 \text{ backwards}}$

Note

The negative answer to the above example indicates that the acceleration is in the opposite direction to the direction chosen as positive (i.e. backwards).

Hence ***deceleration is an acceleration in the backwards direction***, causing the car to slow.

If any answer to a problem is negative (and you are sure of your mathematics!), it means the vector is acting in the opposite direction to the positive direction you chose.

Worksheet 8.3

Acceleration

Student: .... Class: ....

The velocities of two objects were recorded over 10-second time periods, as shown in the examples below.

Example: Gaining and losing speed

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	2	4	6	8	10	12	11	10	9	8

1. (a) Use a spreadsheet application such as Excel to plot a line graph of this data.

(b) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units. Use the printed graph to answer the following questions.

(c) Explain the shape of the graph.

....

....

(d) At the eight-second mark the object's speed begins to decrease. If the object's speed continues to decline at the same rate after ten seconds, at what time will its velocity be zero?

....

....

Example: Falling down

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	9.8	19.6	29.4	39.2		58.8		78.4		98.0

2. (a) Examine the data table and determine the pattern in the data. Use this pattern to complete the table.

(b) Use a spreadsheet application such as Excel to plot a line graph of this data.

(c) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units.

(d) Explain how this graph shows that falling due to gravity is an example of uniform acceleration.

....

....

....

....

....

....

(e) The slope of the graph is equal to the acceleration due to gravity. Determine the slope of the graph in m/s^2.

....

....

Year 10 Science
Physical Sciences
Worksheet - Acceleration

ACCELERATION

Acceleration occurs whenever the velocity of a body changes.

Acceleration is defined as the rate of change of velocity.

For the purpose of examples considered in this text acceleration is considered to be uniform (or constant).

From the definition:

$$\text{Average acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

$$= \frac{\text{final velocity} - \text{initial velocity}}{\text{time}}$$

$$a(av) = \frac{v - u}{t}$$

rearranging gives: $\boxed{v = u + at}$ (assuming uniform acceleration)

where: $v =$ final velocity
$u =$ initial velocity
$a =$ uniform acceleration
$t =$ time interval (measured from zero)

UNITS OF ACCELERATION

As acceleration is the rate of change of velocity then its units are those of velocity divided by time.

Units: $\frac{m\ s^{-1}}{s} = m\ s^{-2}$ (or m/s^2)

TYPE EXAMPLE 10: Determine the acceleration of a body which is accelerated from 2 $m\ s^{-1}$ to 10 $m\ s^{-1}$ in 2.5 s.

$u = 2\ m\ s^{-1}$
$v = 10\ m\ s^{-1}$
$a = ?$
$t = 2.5\ s$

$$v = u + at$$
$$\therefore a = \frac{v - u}{t}$$
$$= \frac{(10 - 2)}{2.5}$$
$$= \frac{8}{2.5}$$
$$= 3.2\ m\ s^{-2}$$

i.e. acceleration is $3.2\ m\ s^{-2}$ (in the direction of the initial motion).

TYPE EXAMPLE 11: What is the initial velocity of a body which attained a velocity of 16 $m\ s^{-1}$ when accelerated at 1.5 $m\ s^{-2}$ for 8 s?

$u = ?$
$v = 16\ m\ s^{-1}$
$a = 1.5\ m\ s^{-2}$
$t = 8\ s$

$$v = u + at$$
$$\therefore u = v - at$$
$$= 16 - (1.5 \times 8)$$
$$= 16 - 12$$
$$= 4\ m\ s^{-1}$$

i.e. the initial velocity is $4\ m\ s^{-1}$ (in the direction of the final velocity)

SET 8: ACCELERATION

1. What is the increase in velocity in each second of a body accelerated at 1.8 m s^{-2}?
2. Calculate the final velocity of a motor car initially travelling at 10 m s^{-1} which is accelerated at 1.4 m s^{-2} for 10 s.
3. A body starts from rest and is accelerated at 2.7 m s^{-2}. What is the velocity after 12 s?
4. A ball rolling down a slope from rest is accelerated at 3.5 m s^{-2}. What is the ball's velocity after 6 s?
5. Find the acceleration of a plane which increases its velocity from 100 m s^{-1} to 200 m s^{-1} in 40 s.
6. What is the average acceleration of a body which is accelerated from rest to 15 m s^{-1} in 5 s?
7. Determine the average acceleration experienced by a body which increases its velocity from 7 m s^{-1} to 34 m s^{-1} in 9 s.
8. A vehicle starting from rest is accelerated at 2.5 m s^{-2}. Determine the time taken to reach a velocity of 5 m s^{-1}.
9. Find the time taken for a body travelling with a velocity of 15 m s^{-1} to reach a velocity of 29 m s^{-1} when accelerated at 0.7 m s^{-2}.
10. A car is accelerated from 10 m s^{-1} to 25 m s^{-1} in 2.5 s. What is the average acceleration?

NEGATIVE ACCELERATION

If a body (travelling in the positive direction) slows down uniformly it can be seen from the equation derived from the definition of acceleration that the value obtained for acceleration will be negative.

That is, as $a = \frac{v - u}{t}$ then if $v < u$, a will be negative.

Whenever the velocity vector and the acceleration vector are in opposite directions the magnitude of the velocity decreases uniformly.

A negative acceleration is also referred to as *RETARDATION* or *DECELERATION.*

TYPE EXAMPLE 12: A car is uniformly slowed down from 26 m s^{-1} to 6 m s^{-1} in a period of 8 s. Calculate the acceleration of the car.

$u = 26$ m s^{-1}
$v = 6$ m s^{-1}
$t = 8$ s
$a = ?$

$$v = u + at$$
$$\therefore a = \frac{v - u}{t} = \frac{6 - 26}{8} = \frac{-20}{8} = -2.5 \text{ m s}^{-2}$$

i.e. acceleration is -2.5 m s^{-2} (or 2.5 m s^{-2} in the opposite direction to the initial motion).

NOTE: When given the rate of slowing down or the retardation assign a negative value to the magnitude (e.g. a body decelerated at 2 m s^{-2} has an acceleration of -2 m s^{-2}).

SET 9: NEGATIVE ACCELERATION

1. A train approaching a railway station at 30 m s^{-1} is brought to rest in 10 s. Determine the retardation.
2. A plane on landing touches down at a speed of 80 m s^{-1}. Determine the acceleration if the plane comes to rest in a time of 8 s.
3. A truck travelling at 36 m s^{-1} is decelerated uniformly at 1.5 m s^{-2}. Calculate the: (a) velocity after 6 s.
   (b) time taken to come to rest.
4. A plane flying at 170 m s^{-1} slows down uniformly at 2 m s^{-2}. What is the plane's velocity after 30 s?
5. How long will it take a vehicle travelling at 30 m s^{-1} to reach a velocity of 12 m s^{-1} if the acceleration is -3 m s^{-2}?
6. A bus travelling at 25 m s^{-1} is decelerated at 2.5 m s^{-2}. Find the time taken for the bus to come to rest.
7. A motor cyclist uniformly decelerates at 3 m s^{-2}. If the initial velocity is 32 m s^{-1}, calculate the time taken to attain a velocity of 14 m s^{-1}.
8. A cyclist brakes suddenly with a uniform acceleration of -4 m s^{-2} and comes to rest in 2.5 s. Determine the initial velocity of the cyclist.

SET 8

1. 1.8 m s^{-1}	**2.** 24 m s^{-1}	**3.** 32.4 m s^{-1}	**4.** 21 m s^{-1}	**5.** 2.5 m s^{-2}
6. 3 m s^{-2}	**7.** 3 m s^{-2}	**8.** 2 s	**9.** 20 s	**10.** 6 m s^{-2}

SET 9

1. 3 m s^{-2}	**2.** -10 m s^{-2}	**3.** (a) 27 m s^{-1}	**(b)** 24 s	**4.** 110 m s^{-1}
5. 6 s	**6.** 10 s	**7.** 6 s	**8.** 10 m s^{-1}	

EQUATIONS OF MOTION

There are three equations which describe the relationship between acceleration, displacement, initial and final velocity and time.

They are as follows.

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

where
- v = final velocity (ms^{-1})
- u = initial velocity (ms^{-1})
- a = acceleration (ms^{-2})
- t = time (s)
- s = displacement (m)

These equations apply only for acceleration occurring in one dimension.

This can be for cars, bikes and other objects moving horizontally across the ground or for objects accelerating vertically under the influence of gravity.

Gravity accelerates all objects down towards the centre of the Earth at 9.80 ms^{-2} at the surface.

Examples

1. A car accelerates uniformly from rest at 3.56 ms-2 and reaches a velocity of 25.0 ms^{-1}. Calculate:

   (a) how far the car moves during this time.
   (b) the time taken for the motion.

   (a)

   $v = 25.0\ ms^{-1}$

   $u = 0.00\ ms^{-1}$

   $a = 3.56\ ms^{-2}$

   $t = ?$

   $s = ?$

   Take forwards as positive.

   $$v^2 = u^2 + 2as$$
   $$Þ\quad (25.0)^2 = 0 + 2(3.56)s$$
   $$s = \frac{(25.0)^2}{2(3.56)}$$
   $$= 87.78\ m$$

   \ <u>Displacement = 87.8 m forwards</u>

(b)

$v = 25.0\ ms^{-1}$ Take forwards as positive.

$u = 0.00\ ms^{-1}$ $v = u + at$

$a = 3.56\ ms^{-2}$ Þ $25.0 = 0 + (3.56)t$

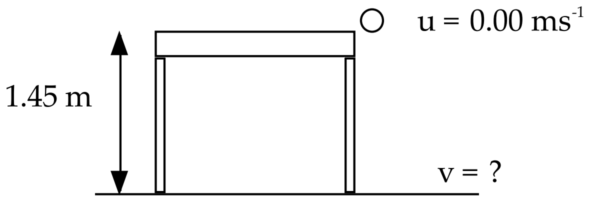
$t = ?$ $s = \dfrac{25.0}{3.56}$

$s = ?$ $= 7.022\ s$

\ Time taken = 7.02 s.

2. A ball rolls off a table 1.45 m high and lands on the floor. Calculate:

(a) the time of flight.
(b) the velocity with which it hits the floor.



Down is positive - in the direction of the acceleration.

(a)

$v = ?$ Take down as positive.

$u = 0.00\ ms^{-1}$ $s = ut + \frac{1}{2}at^2$

$a = 9.80\ ms^{-2}$ Þ $1.45 = 0 + \frac{1}{2}(9.80)t^2$

$t = ?$ $t^2 = \dfrac{1.45}{4.90}$

$s = 1.45\ m$ Þ $t = 0.5440\ s$

\ Time taken = 0.544 s.

(b)

$v = ?$ Take forwards as positive.

$u = 0.00\ ms^{-1}$ $v = u + at$

$a = 9.80\ ms^{-2}$ $= 0 + (9.80)(0.544)$

$t = 0.544\ s$ $= 5.331\ ms^{-1}$

$s = 1.45\ m$

\ Velocity = 5.33 ms^{-1} downwards.

3. A boy kicks a football vertically upwards at 15.0 ms^{-1} from 0.500 m above the ground. The ball moves above the height of a nearby tree and gets caught by a branch 4.00 m above the ground.

(a) Calculate the maximum height reached by the ball.
(b) Determine the velocity of the ball when it is caught by the tree.
(c) How long is the ball in flight?

(a)

Consider only the movement of the ball up to the highest point.

$v = 0.0\ ms^{-1}$ Take down as positive.

$u = -15.0\ ms^{-1}$ $v^2 = u^2 + 2as$

$a = 9.80\ ms^{-2}$ Þ $0 = (-15.0)^2 + 2(9.80)s$

$t = ?$ $s = \dfrac{(-15.0)^2}{2(9.80)}$

$s = ?$ $= -11.48$ m

\ Height = 11.5 m + 0.500 m = 12.0 m above the ground

(b)

Consider the whole movement of the ball .

$v = ?$ Take down as positive.

$u = -15.0\ ms^{-1}$ $v^2 = u^2 + 2as$

$a = 9.80\ ms^{-2}$ $= (-15.0)^2 + 2(9.80)(-3.50)$

$t = ?$ Þ $v = \sqrt{1.564 \times 10^2}$

$s = -3.50$ m $= 12.51\ ms^{-1}$

\ Velocity = 12.5 ms^{-1} down.

(c)

Consider the whole movement of the ball.

$v = 12.5\ ms^{-1}$ Take down as positive.

$u = -15.0\ ms^{-1}$ $v = u + at$

$a = 9.80\ ms^{-2}$ Þ $12.5 = -15.0 + (9.80)t$

$t = ?$ Þ $t = \dfrac{(12.5 + 15.0)}{9.80}$

$s = -3.50$ m $= 2.806$ s

\ Time = 2.81 s.

Year 10 Science
Physical Sciences
Worksheet - Equations of Motion (Extension)

DISPLACEMENT OF UNIFORMLY ACCELERATED BODIES

For any moving body: $v_{av} = \frac{s}{t}$

For a body moving with a uniform acceleration:

$$v_{av} = \frac{u + v}{2}$$

hence:
$$\frac{s}{t} = \frac{u + v}{2}$$
$$= \frac{u + (u + at)}{2} \quad \text{(as } v = u + at\text{)}$$

$$\therefore \boxed{s = ut + \tfrac{1}{2} at^2}$$

TYPE EXAMPLE 13: A vehicle travelling at 10 m s⁻¹ is accelerated at 4 m s⁻². After a period of 5 s, determine the:
(a) displacement
(b) average velocity
(c) final velocity.

(a) $u = 10\ \text{m s}^{-1}$
$a = 4\ \text{m s}^{-2}$
$t = 5\ \text{s}$
$s = ?$

$$\begin{aligned} s &= ut + \tfrac{1}{2} at^2 \\ &= (10 \times 5) + \tfrac{1}{2}(4 \times 5^2) \\ &= 50 + \tfrac{1}{2}(4 \times 25) \\ &= 50 + 50 \\ &= 100\ \text{m} \end{aligned}$$

i.e. the displacement is 100 m (in the direction of the initial motion).

(b) $v_{av} = ?$
$s = 100\ \text{m}$
$t = 5\ \text{s}$

$$\begin{aligned} v_{av} &= \frac{s}{t} \\ &= \frac{100}{5} \\ &= 20\ \text{m s}^{-1} \end{aligned}$$

i.e. the average velocity is $20\ \text{m s}^{-1}$ (in the direction of the initial motion).

(c) $v = ?$
$u = 10\ \text{m s}^{-1}$
$a = 4\ \text{m s}^{-2}$
$t = 5\ \text{s}$

$$\begin{aligned} v &= u + at \\ &= 10 + (4 \times 5) \\ &= 10 + 20 \\ &= 30\ \text{m s}^{-1} \end{aligned}$$

i.e. the final velocity is $30\ \text{m s}^{-1}$ (in the direction of the initial motion).

OR $v = ?$
$u = 10\ m\ s^{-1}$
$v_{av} = 20\ m\ s^{-1}$

$$v_{av} = \frac{u + v}{2}$$
$$\therefore u + v = 2 \times v_{av}$$
$$\therefore v = (2 \times v_{av}) - u$$
$$= (2 \times 20) - 10$$
$$= 40 - 10$$
$$= 30\ m\ s^{-1}$$

i.e. the final velocity is $30\ m\ s^{-1}$ (in the direction of the initial motion).

SET 10: DISPLACEMENT OF UNIFORMLY ACCELERATED BODIES I

1. A car moving with a velocity of $20\ m\ s^{-1}$ is accelerated at $3\ m\ s^{-2}$. After 8 s, determine the:
   (a) displacement.
   (b) average velocity.
   (c) final velocity.
2. What distance will a cyclist travel if when starting from rest the cycle is accelerated at $1.8\ m\ s^{-2}$ for 4 s?
3. A motor boat moving with a velocity of $2\ m\ s^{-1}$ is accelerated at $1\ m\ s^{-2}$ for 6 s. Calculate the:
   (a) displacement.
   (b) final velocity.
   (c) average velocity.
4. A train, initially at rest, leaves a railway station with an acceleration of $0.4\ m\ s^{-2}$ for a period of 50 s. Find the:
   (a) displacement.
   (b) final velocity.
   (c) average velocity.
5. A truck travelling at $15\ m\ s^{-1}$ brakes uniformly at $1.75\ m\ s^{-2}$ for 4 s in coming to rest at a stop sign. At what distance from the sign did the truck commence braking?
6. A ball starting from rest rolls down an inclined plank with an average acceleration of $5\ m\ s^{-2}$. If it takes 2 s to roll off the end, determine the:
   (a) length of the plank.
   (b) final velocity of the ball.
   (c) average velocity of the ball.
7. A bicycle moving at $8\ m\ s^{-1}$ is accelerated at $-1.25\ m\ s^{-2}$ for a 6 s period. Determine the:
   (a) displacement.
   (b) velocity at the end of this period.
8. A body with a speed of $50\ m\ s^{-1}$ is subjected to a uniform retardation of $10\ m\ s^{-2}$ for 4 s. Determine the:
   (a) distance travelled.
   (b) average speed.
   (c) final speed.
9. A plane can effect take-off with a velocity of $72\ m\ s^{-1}$. If the plane can accelerate along the runway at $4\ m\ s^{-2}$, determine the:
   (a) time taken to reach a speed of $72\ m\ s^{-1}$
   (b) minimum length of runway to enable a safe take-off.
10. A person starting from rest takes 6 s to reach a maximum velocity down a ski slope. Assuming a constant acceleration of $5\ m\ s^{-2}$ find the:
   (a) distance travelled in 6 s
   (b) maximum velocity attained by the skier.

SET 10

1. (a) 256 m (b) 32 m s^{-1} (c) 44 m s^{-1} 2. 14.4 m
3. (a) 30 m (b) 8 m s^{-1} (c) 5 m s^{-1} 4. (a) 500 m (b) 20 m s^{-1} (c) 10 m s^{-1}
5. 46 m 6. (a) 10 m (b) 10 m s^{-1} (c) 5 m s^{-1} 7. (a) 25.5 m (b) 0.5 m s^{-1}
8. (a) 120 m (b) 30 m s^{-1} (c) 10 m s^{-1} 9. (a) 18 s (b) 648 m
10. (a) 90 m (b) 30 m s^{-1}

ANOTHER EQUATION OF MOTION

The equation, $v^2 = u^2 + 2as$ is often used to solve in one step problems which require the use of both of the other two equations of motion, $v = u + at$, and $s = ut + ½ at^2$.

GRAVITATIONAL ACCELERATION

All freely falling bodies close to the earth's surface are accelerated at the same rate (i.e. they fall the same distance in the same time). The acceleration due to gravity is often referred to as g.

Acceleration due to gravity acts in a vertically downward direction and has a value of approximately 10 m s^{-2}.

The previously established equations of motion for uniformly accelerated bodies may be used for freely falling bodies.

$$v = u + gt$$
$$s = ut + ½ gt^2$$

where: g = acceleration due to gravity
= 10 m s^{-2} (vertically downward)

NOTE: 1. The effects of frictional air resistance are ignored and are assumed to be negligible.
2. At ground level g has a value of about 9.8 m s^{-2}. For ease of calculation the rounded off value of 10 m s^{-2} is used.

TYPE EXAMPLE 14: A stone falls off the edge of a cliff. If it takes 6 s to strike the ground below, determine the:

(a) velocity with which the stone strikes the ground.

(b) height of the cliff.

(a)

$u = 0$
$t = 6\ s$
$g = 10\ m\ s^{-2}$
$v = ?$

$$
\begin{aligned}
v &= u + gt \\
&= 0 + (10 \times 6) \\
&= 60\ m\ s^{-1}
\end{aligned}
$$

i.e. the stone strikes the ground at 60 m s^{-1} (downwards).

(b)

$u = 0$
$t = 6\ s$
$g = 10\ m\ s^{-2}$
$s = ?$

$$
\begin{aligned}
s &= ut + \tfrac{1}{2} gt^2 \\
&= 0 + \tfrac{1}{2}(10 \times 6^2) \\
&= \tfrac{1}{2}(10 \times 36) \\
&= 180\ m
\end{aligned}
$$

i.e. as the displacement is 180 m (downwards) the height of the cliff is 180 m.

SET 12: GRAVITATIONAL ACCELERATION I

NOTE: Use acceleration due to gravity, $g = 10\ m\ s^{-2}$.

1. A stone drops freely from rest. Determine after 4 s the:
   (a) stone's final velocity
   (b) displacement.
2. A can of paint falls from the top of a painter's ladder.
   If it takes 1.2 s for the can to strike the ground below, find the:
   (a) final velocity of the can
   (b) length of the ladder (assumed to be vertical).
3. A cricket ball is hit vertically into the air. The ball at its maximum height is momentarily at rest and from this position takes 2.5 s to strike the ground. Calculate the height to which the ball was hit.
4. A stone is dropped from rest into a well. Find the depth of the well if it takes 0.9 s for the stone to strike the bottom.
5. A ball is thrown vertically downwards at 10 m s^{-1}. How far does the ball travel in 3 s?
6. An object drops (from rest) out of a window of a tall building. How far does the object fall:
   (a) in 2 s?
   (b) in 3 s?
   (c) in the 3rd second?

SET 12

1. (a) 40 m s^{-1} **(b)** 80 m **2.** (a) 12 m s^{-1} **(b)** 7.2 m **3.** 31.25 m
4. 4.05 m **5.** 75 m **6.** (a) 20 m **(b)** 45 m **(c)** 25 m

ACCELERATION DUE TO GRAVITY (EXTENDED)

When a projectile has an initial velocity upwards the sign convention *MUST* be used as 'g' is *ALWAYS* directed vertically downwards. As it rises the projectile slows until at its maximum height its velocity is *MOMENTARILY* zero. This occurs because the initial velocity and acceleration vectors are in opposite directions. From its maximum height the projectile falls with a velocity increasing in the downward direction until, at the moment it returns to the position from which it was projected, it is of the same magnitude as when it was projected vertically but in the *OPPOSITE* direction. This occurs because the final velocity and acceleration vectors are in the same direction.

For the entire journey g is constant (and equal to 10 m s^{-2} downwards).

Generally downward vectors are designated a *POSITIVE* sign.

TYPE EXAMPLE 15: A stone is thrown vertically upward with a velocity of 35 m s^{-1}. Find the velocity of the stone after

(a) 2 s

(b) 4 s.

(a)

$g = +10 \text{ m s}^{-2}$
$u = -35 \text{ m s}^{-1}$
$t = 2 \text{ s}$
$v = ?$

$$\begin{aligned} v &= u + gt \\ &= -35 + (10 \times 2) \\ &= -35 + 20 \\ &= -15 \text{ m s}^{-1} \end{aligned}$$

i.e. the velocity after 2 s is 15 m s^{-1} UPWARDS.

(b)

$g = +10 \text{ m s}^{-2}$
$u = -35 \text{ m s}^{-1}$
$t = 4 \text{ s}$
$v = ?$

$$\begin{aligned} v &= u + gt \\ &= -35 + (10 \times 4) \\ &= -35 + 40 \\ &= 5 \text{ m s}^{-1} \end{aligned}$$

i.e. the velocity after 4 s is 5 m s^{-1} DOWNWARDS.

To determine the maximum height reached by a projectile during its flight the following may assist:

1. *at maximum height, the velocity is momentarily zero.*
2. *the time taken to reach maximum height is the same as the time taken to return from the position of maximum height.*

TYPE EXAMPLE 16: An arrow is fired vertically upwards with a velocity of 80 m s^{-1}.

Determine the: (a) time taken to reach maximum height.
(b) maximum height reached by the arrow.

(a)

$g = +10 \text{ m s}^{-2}$
$u = -80 \text{ m s}^{-1}$
$v = 0$
$t = ?$

$$v = u + gt$$
$$\therefore gt = v - u$$
$$\therefore t = \frac{v-u}{g}$$
$$= \frac{0-(-80)}{10}$$
$$= 8 \text{ s}$$

i.e. the time taken to reach maximum height is 8 s.

(b) *Consider the arrow falling from its position of maximum height.*

The time taken to reach maximum height is equal to the time taken for the arrow to return $\therefore t = 8$ s .

$g = 10 \text{ m s}^{-2}$
$t = 8 \text{ s}$
$u = 0$
$s = ?$

$$s = ut + \tfrac{1}{2}gt^2$$
$$= 0 + \tfrac{1}{2}(10 \times 8^2)$$
$$= \tfrac{1}{2}(10 \times 64)$$
$$= 320 \text{ m}$$

i.e. As the displacement of the falling arrow is 320 m (DOWNWARDS) the maximum height reached by the arrow is 320 m.

OR

Consider the arrow rising to its position of maximum height.

$s = ?$
$g = 10 \text{ m s}^{-2}$
$u = -80 \text{ m s}^{-1}$
$t = 8$ s (from part a)

$$s = ut + \tfrac{1}{2}gt^2$$
$$= (-80) \times 8 + \tfrac{1}{2}(10 \times 8^2)$$
$$= -640 + 320$$
$$= -320 \text{ m}$$

i.e. As the displacement is 320 m (UPWARDS) the maximum height reached by the arrow is 320 m.

SET 13: GRAVITATIONAL ACCELERATION II

NOTE: Use acceleration due to gravity, $g = 10\ m\ s^{-2}$.

1. A stone is thrown vertically downwards at 10 m s^{-1}. After 2.5 s, find the:
   (a) velocity.
   (b) displacement.
2. A ball is thrown vertically downwards into a well with a velocity of 15 m s^{-1}. If the time taken to strike the water is 1 s find the:
   (a) velocity with which the stone strikes the water.
   (b) depth of the well to the water level.
3. A bullet is fired vertically upwards at 600 m s^{-1}.
   What is the velocity of the bullet after:
   (a) 20 s?
   (b) 100 s?
4. A ball is thrown vertically upwards at 13 m s^{-1}. What is its velocity after 1.6 s?
5. A bullet is fired vertically upwards at 600 m s^{-1}. Determine the:
   (a) time taken for the bullet to reach the maximum height
   (b) height reached by the bullet.
6. A stone is thrown vertically upward at 16 m s^{-1}. Calculate the:
   (a) time taken to reach maximum height
   (b) height reached by the stone.
7. A brick falls from rest off the top of a 45 m tall construction site. How long does it take to strike the ground?
8. How long would it take for a parcel to strike the ground if dropped from rest out of a helicopter stationary at a height of 80 m?
9. A stone is dropped from rest. Calculate the:
   (a) time taken to fall 12.8 m
   (b) velocity at this displacement.
10. An object is thrown vertically upward at 17.2 m s^{-1}. How long will it take for the object to return to the ground?

SET 13

1. **(a)** 35 m s^{-1} **(b)** 56.25 m **2.** **(a)** 25 m s^{-1} **(b)** 20 m
3. **(a)** 400 m s^{-1} upwards **(b)** 400 m s^{-1} downwards
4. 3 m s^{-1} downwards **5.** **(a)** 60 s **(b)** 18 000 m **6.** **(a)** 1.6 s **(b)** 12.8 m
7. 3 s **8.** 4 s **9.** **(a)** 1.6 s **(b)** 16 m s^{-1} **10.** 3.4 s

FORCES

Forces can cause the following changes to objects they act on. They can:

- cause an object to accelerate—that is, to increase its velocity.
- cause an object to decelerate—that is, to decrease its velocity.
- cause an object to change its direction.
- cause a change of shape.

In order for a force to do these things, two conditions must be met.

1. ***The force must be unbalanced.***

   In order for a force to cause a change of velocity, direction or shape there must be an unbalanced force in the direction of the change.

   If the force is balanced by another force, the forces cancel each other out because they are acting in opposite directions.

This box will experience a change in its motion because the forces acting are not balanced. The larger force to the left will cause the box to accelerate to the left.

Fig 2.2.2

2. ***The force must be external.***

   Can you move a car by sitting inside it and pushing on the steering wheel? Of course, the answer is no.

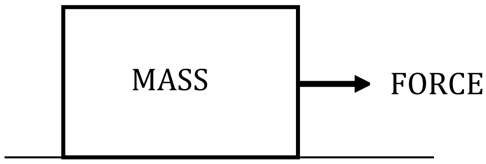
   If you wanted to push a car, you would have to stand outside the car and push it.

To fully understand how forces can do these things, one needs to understand Newton's Three Laws of Motion.

NEWTON'S FIRST LAW

A body at rest or moving at constant velocity will continue to do so unless acted upon by an external unbalanced force.

Example



The force needed to move an object appears to be dependent on the mass.

The larger the mass, the greater the force required to move the object.

Newton's First Law is often referred to as the law of ***inertia***.

Inertia is considered to be the property of a body which, when still, makes it hard to move and, when moving, keeps it going.

This property seems to be determined by mass - ***the greater the mass, the greater the inertia***.

Examples

1. A train has a lot of inertia due to its large size. It takes a great deal of force to start it moving (if it's stationary) or stop it (if it's moving).

2. Consider a passenger in a car moving at 60.0 km/h. If the car brakes suddenly, the passenger will continue to move forward at 60.0 km/h (due to inertia), until the seatbelt exerts a force on them to slow them down.

Fig 2.2.6 A passenger in a car demonstrates Newton's First Law when the car slows down rapidly.

3. If a car accelerates quickly from a standing start, a passenger is thrown back in their seat. They want to stay stationary (due to inertia) and the car moves forwards under them.

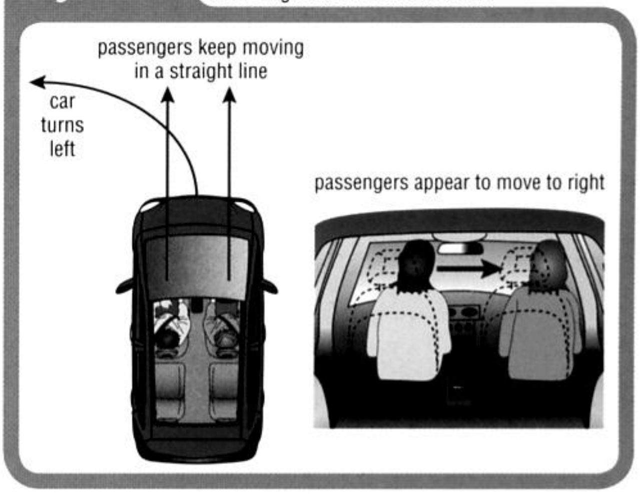
When a car accelerates you feel as if you are being pushed back into your seat. Fig 2.2.8

4. When a car turns a corner, the passengers have inertia that wants to keep them moving straight ahead. The car moves under them and the passengers feel as if they are moving (leaning) in the opposite direction to the turn.

Because of this, people assume a ***centrifugal force*** is acting on the passengers pushing them away from the corner.

THIS IS FALSE - THE "FORCE" DOESN'T EXIST!!

Fig 2.2.9 You feel as though you are being pushed to the right as the car turns left.



MORE ABOUT FORCES

Unbalanced Forces

In the diagram at right, the bike experiences an unbalanced force of 20 Newtons in a forward direction.

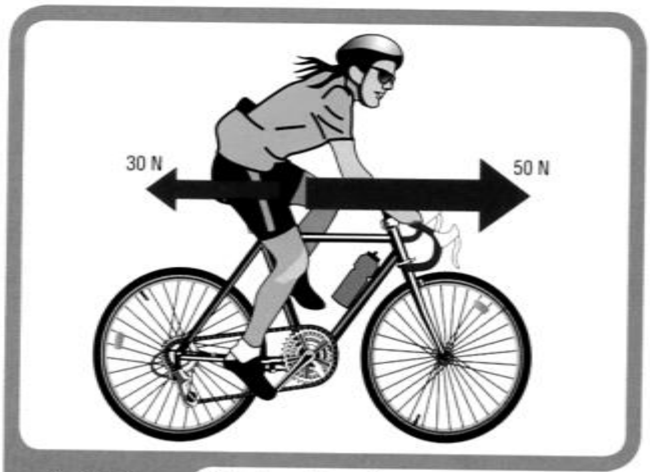


Fig 2.3.1 This bike will accelerate.

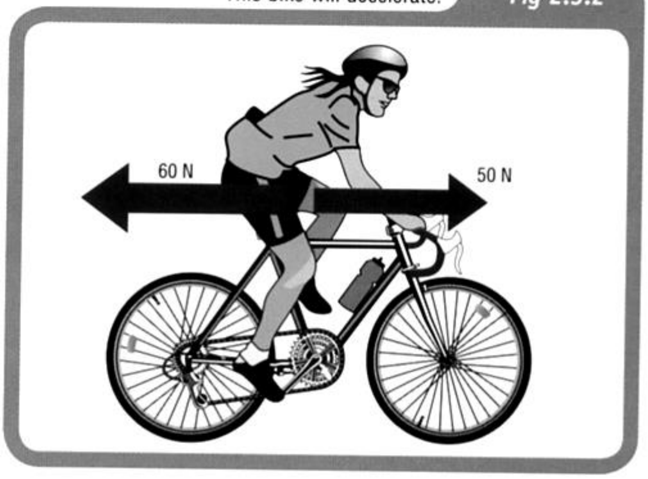
The bike will therefore accelerate in a forward direction because of the effect of this unbalanced forward force.

To calculate the resultant force, ***take forwards as positive***.

=> The force backwards is negative.

∴ $\underline{\Sigma F = 50 - 30 = 20 \text{ N forwards}}$.

This bike will decelerate. Fig 2.3.2



However, if the girl is not pedalling hard enough to overcome all the friction, the unbalanced force acting on the bike is 10 newtons in the opposite direction from the direction that the bike is moving.

The bike will therefore not be able to maintain a constant velocity but will decelerate.

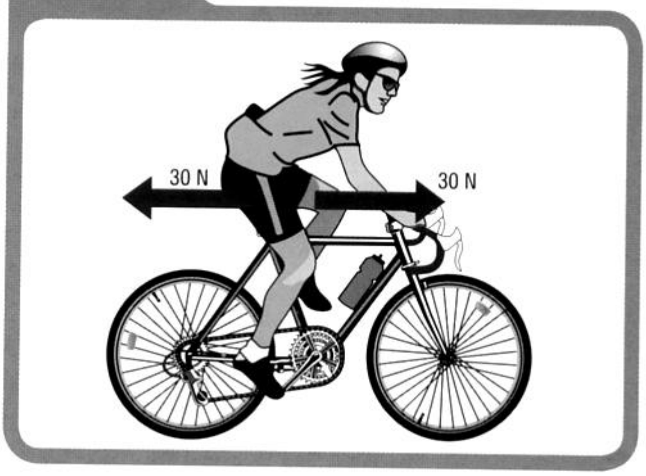
Take forwards as positive.

$\Sigma F = 50 - 60 = -10$ N

the negative sign means the direction is

∴ $\underline{\Sigma F = 10 \text{ N backwards.}}$

Fig 2.3.3 This bike will continue moving in a straight line at a constant speed.



If the girl's forward pedalling force is exactly the same size as the frictional force in the opposite direction, there is no unbalanced force acting on the bike.

The bike will not accelerate or decelerate but will continue in a straight line with a constant speed.

∴ $\underline{\Sigma F = 0}$

Worksheet 8.4

Inertia and motion

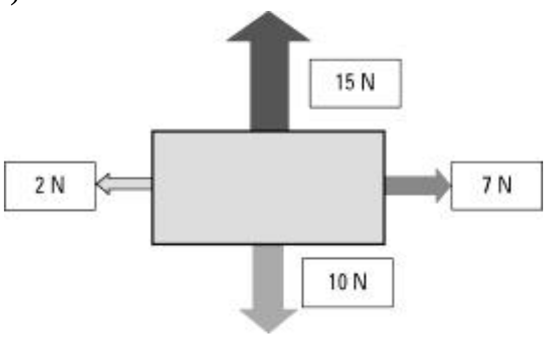
Student: .... Class: ....

1. State Newton's First Law of Motion.

2. Consider the following examples of forces acting on a body. In each case find the net (unbalanced) force and describe the resulting motion of the object. Assume north is up the page.

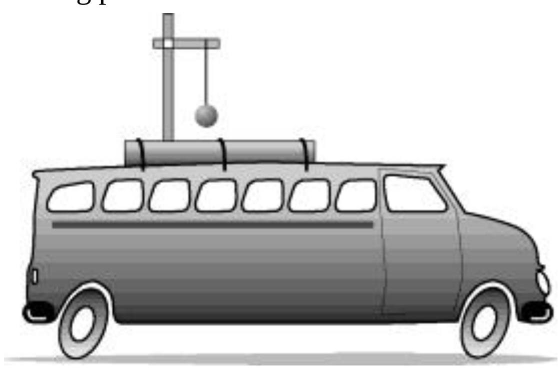
(a)

(b)



3. A 500 kg box is placed on the tray of a utility truck but is not secured by ropes. The ute drives along a highway, then the driver has to brake suddenly to avoid a wombat on the road. Use the concept of inertia to explain the danger to the driver.

4. A 1.0 kg pendulum is mounted on the roof of a vehicle as shown in the diagram.



Describe the motion of the pendulum for each example.

(a) The vehicle accelerates from rest.

(b) The vehicle brakes from a speed of 60 km/h.

NEWTON'S SECOND LAW

> An external unbalanced force will cause an object to accelerate in the direction of the resultant force.

This is represented by: $\boxed{\Sigma F = ma}$

where ΣF = the resultant unbalanced force acting (N)
m = mass of the object (kg)
a = acceleration (ms^{-2})

Examples

1. What would be the acceleration of a skateboarder and her skateboard if their combined mass of 55.0 kg was acted on by an accelerating force of 60.0 N?

Take forwards as positive.

SF = 60.0 N

m = 55.0 kg

a = ?

SF = ma

Þ $a = \frac{SF}{m}$

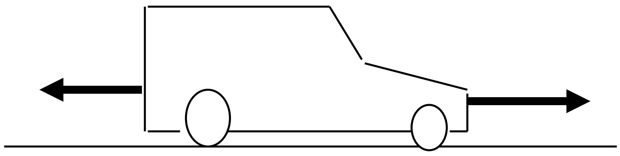
$= \frac{60.0}{55.0}$

$= 1.091\ m/s^2$

\ <u>Acceleration = 1.09 m/s^2 forwards</u>

2. A car of mass 1100 kg is accelerated by a motor that provides a force of 3100 N. A frictional force of 850 N opposes the motion of the car. Calculate the acceleration of the car across the ground.

$F_{friction}$ = 850 N



F_{engine} = 3100 N

Take forwards as positive.

$$\Sigma F = 3100 - 850$$
$$= 2250 \text{ N}$$

ΣF = 2250 N

m = 1100 kg

a = ?

$$\Sigma F = ma$$
$$Þ \quad a = \frac{\Sigma F}{m}$$
$$= \frac{2250}{1100}$$
$$= 2.045 \text{ m/s}^2$$

\ <u>Acceleration = 2.04 ms^{-2} forwards</u>

Gravity and Weight

Most objects fall back to Earth due to gravity, one of the fundamental forces in the universe.

According to Isaac Newton, gravity is due to the attraction between two masses.

i.e. There is an attraction between you and the person next to you, your pens and pencils, the Earth, etc.

The attraction to the Earth is the strongest since its mass is so large (5.98×10^{24} kg).

Two terms need to be clearly distinguished.

Mass - amount of matter in an object (measured in kg).

Weight - force due to gravity on an object (measured in Newtons - N).

The mass of an object does not change throughout the universe while the weight will constantly change as it is moved from place to place.

As seen previously, the acceleration due to gravity on the earth's surface is:

$$g = 9.80 \text{ m/s}^2$$

The force due to weight (F_w) is given by:

$$\boxed{\mathbf{F_w = mg}}$$

where g = the acceleration due to gravity (m/s^2)

Example

The gravitational field on Jupiter is 2.50 times greater than that on Earth. If it was possible for a person of 80.0 kg to stand on Jupiter's surface, calculate their weight.

$F_w = ?$

$m = 80.0 \text{ kg}$

$g = (2.50 \times 9.80) \text{ m/s}^2$

$$\begin{aligned} F_w &= mg \\ &= (80.0)(2.50 \times 9.80) \\ &= \frac{60.0}{55.0} \\ &= 1960 \text{ N} \end{aligned}$$

\ <u>Weight = 1960 N</u>

Force and gravity

Student: .... Class: ....

Weight

1. The weight of an object is the force acting on the object in a gravitational field. The stronger the gravitational field, the greater the weight of a body. Weight is measured in the units of newtons (N)

The weight (W) of a body of mass (m) in a gravitational field (g = acceleration due to gravity) is given by the formula:

$$W = mg$$

On the Earth's surface 'g' typically has a value of 9.8 m/s^2 but this varies with altitude and latitude. On other planets, the value of 'g' also changes.

Complete the table using the above formula.

Mass (kg)	Gravity (m/s^2)	Weight (N)
20	9.8	
30		286.5
	9.8	400.0

Spreadsheets and graphing

2. A spreadsheet program such as Excel is a useful computer application for tabulating data and plotting graphs. In this exercise you will practise using a spreadsheet application.

The following mass and weight data was collected from the planet Mars. The data is presented as you would set it out in the spreadsheet application.

(a) Open the spreadsheet application and enter the data as tabulated below.
(b) Select *Insert* from the menu bar.
(c) Select *Scatter* as the chart type. Select 'Scatter with straight lines and markers'.
(d) The *Design* and *Layout* tabs can be used to insert a chart title and then each axis title and unit.
(e) Select the gridlines you would like to show.

	A	B
1	**Mass (kg)**	**Weight (N)**
2	0.0	0.0
3	0.5	1.9
4	1.0	3.8
5	1.5	5.7
6	2.0	7.6
7	2.5	9.5
8	3.0	11.4

Use your chart to answer the following questions.

(f) Estimate the weight of a 1.3-kg body on the surface of Mars. ....
(g) Estimate the mass of an object that weighs 10.0 N on Mars. ....
(h) Suggest what information the slope of the graph may provide.

(*Clue:* The formula $W = mg$ can be rearranged as $g = \frac{W}{m}$.)

....

....

Newton's Second Law

Student: .... Class: ....

1. State Newton's Second Law of Motion.

....

....

....

2. Use the formula $F = ma$ to complete the following table

Force (F) (newtons)	Mass (m)	Acceleration (a)
100	40 kg	
2	200 g	
70		3.5 m/s^2
100		10 cm/s^2

3. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.

(a) Calculate the net force acting on the body.

....

....

....

....

(b) Calculate the acceleration of the body.

....

....

....

....

4. The photo shows a drag car coming to a stop at the end of a race.

Photo: Claudia Schnepf / 123RF

(a) To come to a complete stop, the drag car negatively accelerates. Identify another term for 'negative acceleration'. ....

(b) Identify the force that brings the drag car to a stop. ....

Year 10 Science
Physicla Sciences
Worksheet - Newton's Laws

NEWTON'S FIRST LAW OF MOTION

"An object continues in its state of rest or uniform motion in a straight line unless acted on by a net external force".

This law is also known as the *law of inertia. Inertia is the tendency of a body to remain in its state of motion.*

The greater the mass of a body the greater is its inertia.

FORCE

A force may be defined as that which produces or tends to produce a change in the state of rest or motion of a body.

NEWTON'S SECOND LAW OF MOTION

"When a net force acts on a body the body is accelerated in the direction of the force. The acceleration is directly proportional to the force and inversely proportional to the mass of the body".

Mathematically this law can be expressed as:

$$a \propto \frac{F}{m}$$

or $F = k\,m\,a$ where
F = net force
k = proportionality constant
m = mass of the body
a = acceleration produced

When F, m and a are expressed in SI units $k = 1$ and the law may be written in its common form:

$$\boxed{F = m\,a}$$

UNITS OF FORCE

The unit of force is the *newton (N)* and is defined such that *a net force of 1 N acting on a mass of 1 kg produces an acceleration of 1 m s⁻²*.

$$\begin{aligned} F &= m\,a \\ &= (1\ \text{kg})(1\ \text{m s}^{-2}) \\ \therefore 1\ \text{N} &= 1\ \text{kg m s}^{-2} \end{aligned}$$

TYPE EXAMPLE 21: What force acting on a mass of 8 kg will produce an acceleration of 1.5 m s⁻²?

$F = ?$
$m = 8$ kg
$a = 1.5\ \text{m s}^{-2}$

$$\begin{aligned} F &= ma \\ &= 8 \times 1.5 \\ &= 12\ \text{N} \end{aligned}$$

i.e. a force of 12 N is required (in the direction of the acceleration).

TYPE EXAMPLE 22: A force of 50 N is applied to a mass of 12.5 kg. What is the acceleration produced?

$$F = 50\text{ N} \qquad F = ma$$
$$m = 12.5\text{ kg} \qquad \therefore a = \frac{F}{m}$$
$$a = ? \qquad = \frac{50}{12.5}$$
$$= 4\text{ m s}^{-2}$$

i.e. an acceleration of 4 m s^{-2} is produced (in the direction of the force).

SET 15: NEWTON'S SECOND LAW OF MOTION I

1. A mass of 6 kg is accelerated at 3.5 m s^{-2}. What net force is applied?
2. Determine the magnitude of the force which when acting on a mass of 2.5 kg produces an acceleration of 5 m s^{-2}.
3. What force must act on a car of mass 2 000 kg to produce an acceleration of 1.6 m s^{-2}?
4. Find the acceleration produced when a force of 200 N acts on a body of mass 40 kg.
5. A force of 36 N acts on a 24 kg mass. What acceleration is produced?
6. What acceleration is produced when a net force of 16 N acts on a mass of 48 kg?
7. Determine the acceleration resulting when a force of 625 N acts on a body of mass 125 kg?
8. A force of 72 N produces in a body an acceleration of 1.2 m s^{-2}. What is the mass of the body?
9. What is the mass of a car which is accelerated at 2 m s^{-2} when acted on by a force of 2 800 N?
10. A force of 3 N is applied to a block on a smooth horizontal table and produces an acceleration of 11 m s^{-2}. What is the mass of the block?

SET 15

1. 21 N **2.** 12.5 N **3.** 3 200 N **4.** 5 m s^{-2} **5.** 1.5 m s^{-2}
6. 0.33 m s^{-2} **7.** 5 m s^{-2} **8.** 60 kg **9.** 1 400 kg **10.** 0.27 kg

WEIGHT AND MASS

Mass refers to the quantity of matter in a body and is measured in kilograms (kg).

Weight is the earth's gravitational force of attraction on a body and is directed towards the centre of the earth. As weight is a force it is measured in *newtons (N)*.

Freely falling bodies are accelerated at 10 m s^{-2} (g) near the earth's surface.

A mass of 1 kg falling freely near the surface of the earth will be accelerated at 10 m s^{-2}. The gravitational force of attraction producing this acceleration can be calculated by:

$$\begin{aligned} F &= ma \\ &= (1\text{ kg})(10\text{ m s}^{-2}) \\ &= 10\text{ kg m s}^{-2} \\ &= 10\text{ N} \end{aligned}$$

From this it can be seen that the gravitational force of attraction on a 1 kg mass is 10 N and from the definition of weight, the weight of a mass of 1 kg is 10 N downward.

As all bodies are accelerated at g near the earth's surface the weight of a body can be determined from:

$$F = ma$$

$$\boxed{F_w = mg}$$

where: F_w = weight (N)
m = mass (kg)
g = acceleration due to gravity

TYPE EXAMPLE 25: Determine the weight of a 6 kg mass.

$m = 6$ kg
$g = 10$ m s^{-2}
$F_w = ?$

$$\begin{aligned} F_w &= mg \\ &= 6 \times 10 \\ &= 60\text{ N} \end{aligned}$$

i.e. the weight of a 6 kg mass is 60 N downward.

SET 17: WEIGHT

Use $g = 10$ m s^{-2} for acceleration due to gravity near the earth's surface.

1. What is the weight of a:
   (a) 52 kg mass? (b) 2.6 kg mass?
2. Determine the mass of bodies with the weights:
   (a) 350 N (b) 5 N
3. Find the weight of a student of mass 48 kg.
4. What is the weight of a 1 000 kg motor car?
5. What is the mass of a crate which weighs 1 200 N?
6. A vehicle of mass 800 kg has a horizontal force equal to its weight applied to it. Find the:
   (a) weight of the vehicle
   (b) acceleration produced.
7. A body of mass 4 kg at rest is acted upon by a force equal to half its weight. Determine after 3 s, the:
   (a) velocity
   (b) displacement.
8. A body is suspended on a spring balance and shows a weight of 20 N. This same balance and body on the surface of the moon would show 3.2 N. What is the acceleration due to gravity on the surface of the moon?

SET 17

1. **(a)** 520 N **(b)** 26 N **2.** **(a)** 35 kg **(b)** 0.5 kg **3.** 480 N **4.** 10 000 N
5. 120 kg **6.** **(a)** 8 000 N **(b)** 10 m s^{-2} **7.** **(a)** 15 m s^{-1} **(b)** 22.5 m
8. 1.6 m s^{-2}

Car Safety Features

Collisions

The ***first collision*** is when the vehicles and/or object collide and the energy of movement (kinetic energy) causes damage to the car(s).

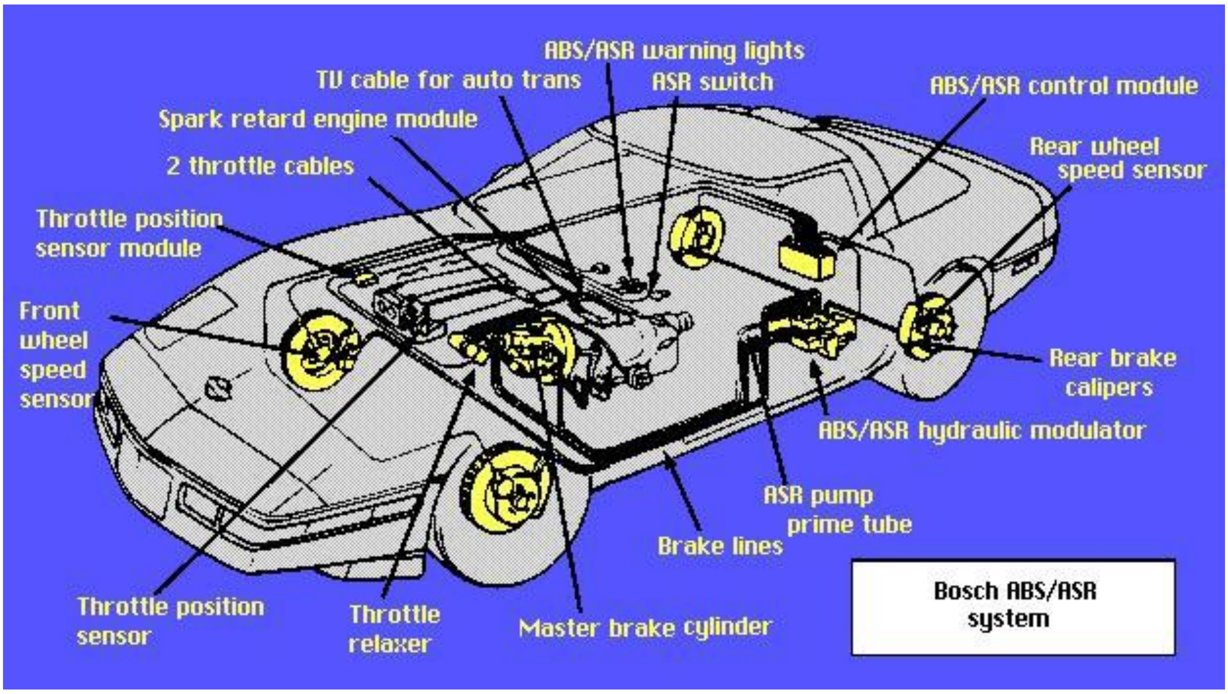
The ***second collision*** is when the passengers in the car collide with the inside of the car they they are travelling in.

Car safety features have been designed to either prevent the first collision occurring, or if it does, then to prevent the second collision occurring.

Primary Safety Features

These are the features of a car that help prevent an accident happening.

- Good brakes.
- Good tyres.
- Good vision from the car.
- Good steering.



Secondary Safety Features

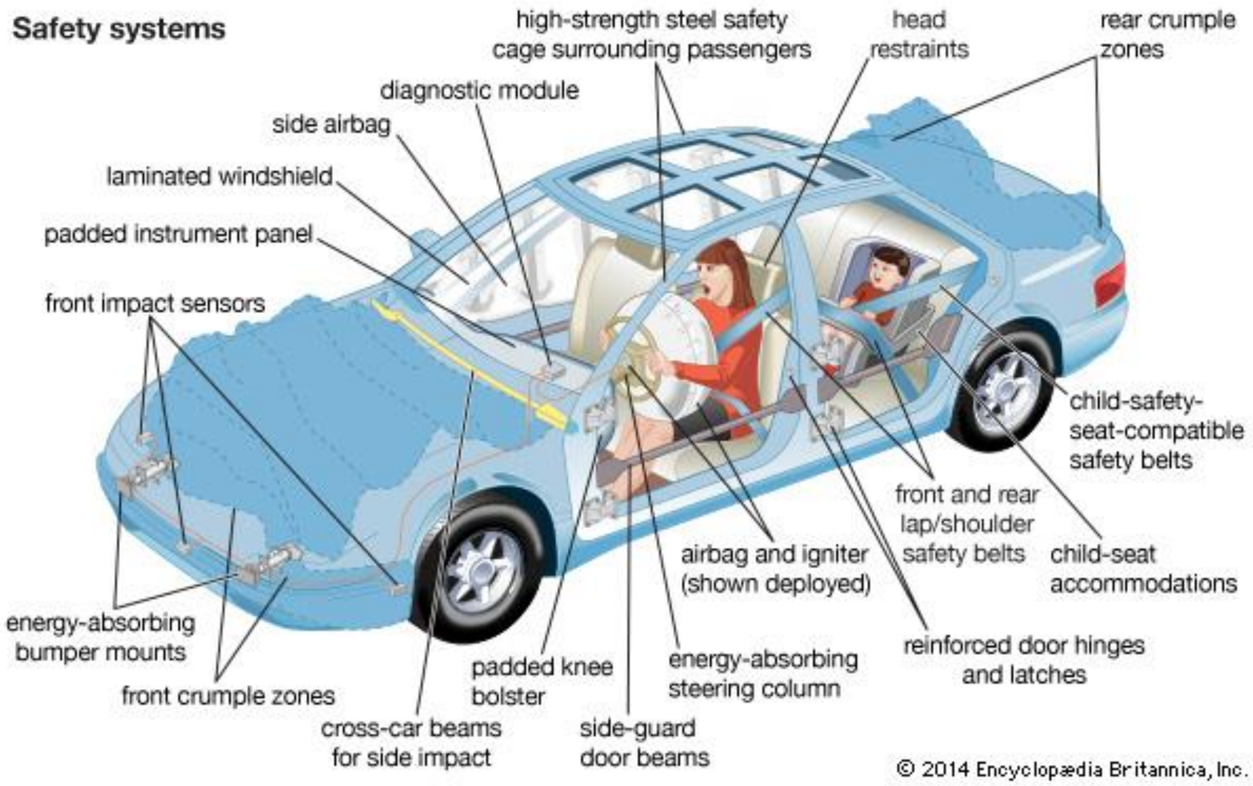
These are the features of a car that help to prevent or reduce injury to its occupants if it is involved in an accident.

They reduce injury by one of the following methods.

- Stopping the occupants moving around in the vehicle during an accident.
- Preventing the vehicle collapsing in onto its occupants.
- Bring the vehicle to rest more slowly or bring the occupant to rest more slowly.

The main secondary safety features in a car are:

- seat belts.
- air bags.
- crumple zones.
- padded and smooth dashboards and other surfaces.



Newton's Second Law and Safety Features

The primary aim of secondary safety features is to make the car or its occupants slow down over a longer time. Why is this desirable?

The mass of the car (or passenger) and the change in velocity ($\Delta v = v - u$) remain constant.

Newton's Second Law can be written as:

$$F = ma$$

$$= m\left(\frac{v-u}{t}\right)$$

$$= \frac{mDv}{t}$$

Since $m\Delta v$ is constant and doesn't change, the size of the force (F) is inversely proportional to the time to stop moving (t).

i.e. $F \;\mu\; \frac{1}{t}$

If ***t*** is made large by slowing the car (or passenger) over a longer time, the force (F) experienced is much smaller.

Hence the passengers are protected.

NEWTON'S THIRD LAW

For every action there is an equal and opposite reaction.

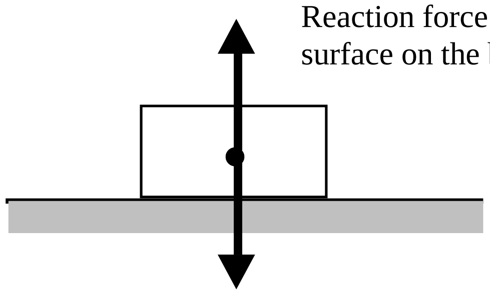
Newton realised that forces occur in pairs and act on different objects.

e.g. Consider a block sitting at rest on a table.

The force due to gravity (weight) acts downwards on the block. Thus the block exerts a force onto the table.

The table exerts an equal and opposite force to support it. The table exerts a force onto the block.

Reaction force of the surface on the block.

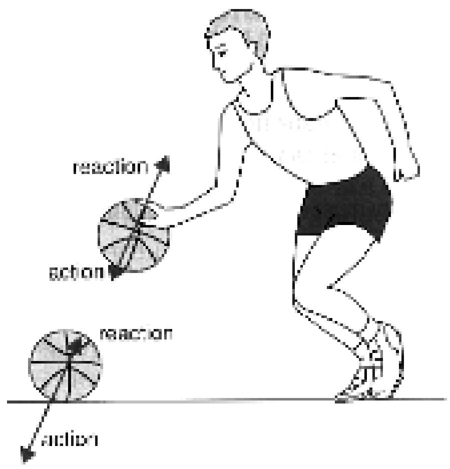


Action force of weight on the

Examples

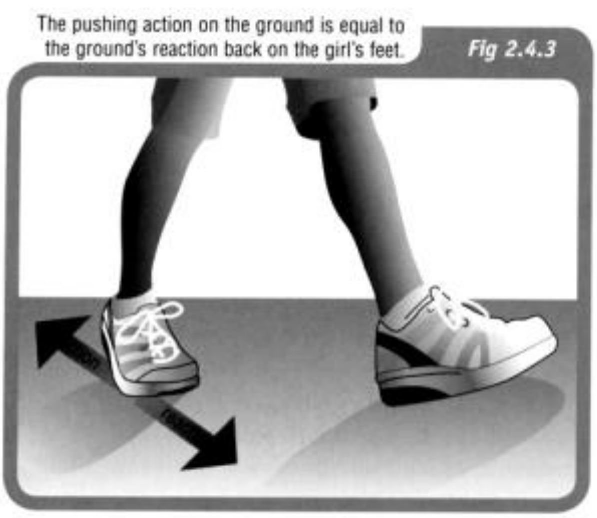
1. The player pushes on the ball (action) and the ball pushes back equally on his hand (reaction).

   When the ball hits the ground, the ball pushes on the ground (action) while the ground pushes back equally (reaction).



2. The girl is able to walk because friction allows her to push backwards on the ground.

   The ground in turn pushes back on her feet, by the same force but in the opposite direction.



The pushing action on the ground is equal to the ground's reaction back on the girl's feet.

Fig 2.4.3

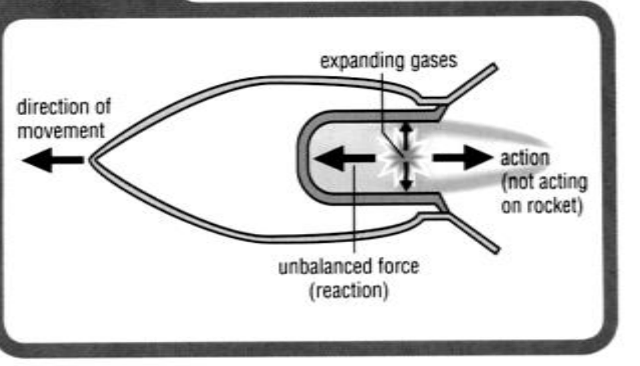
3. The diagram shows the reaction chamber for a rocket.

   As the fuel burns, the expanding gases push in all directions.

   The gases rush out the exhaust outlet (action), causing a push on the rocket (reaction) that moves it forwards.

   This unbalanced force is called ***thrust***.

Fig 2.4.8 The exhaust leaving the rocket creates an unbalanced forwards force in the rocket.



Worksheet 8.7

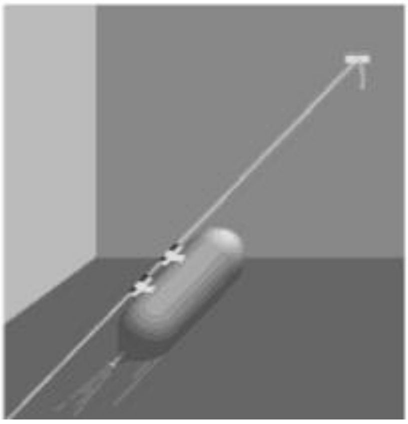
Newton's Third Law

Student: .... Class: ....

1. State Newton's Third Law of Motion.

....
....
....

2. The following diagram shows a balloon rocket built by students to investigate motion. The balloon is mounted on a horizontal string using a length of drinking straw and masking tape.



The inflated balloon is then released.

(a) Where along the string will the balloon be moving at the greatest speed?

....
....
....

(b) Use Newton's Third Law to explain why the balloon is propelled forward along the string.

....
....
....
....
....

3. The photograph below shows a person pushing against a wall.

Photo: Wavebreak Media Ltd / 123RF

The man is not moving even though he is pushing forward against the wall.

(a) What is the net force acting on the man?

....

(b) If the man pushes with a force of 200 N on the wall, what is the force that the wall exerts back on him?

....
....

(c) The man is pushing back on the floor with his feet with a force 200 N. With what force does the floor push back on the man?

....
....

(d) Explain what would happen if he tried to do this on a very slippery floor.

....
....
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....

Worksheet 8.8

Forces, energy and motion: Summary

Student: .... Class: ....

Complete the statements below to compile a summary of this unit. The missing words can be found in the word list below.

1. Speed is a measure of the …………… at which an object moves over a distance.
2. …………… speed can be calculated by dividing the distance travelled by the time taken.
3. …………… is a measure of the rate of change in position. In order to describe the velocity fully, the …………… of the change in position must be stated.
4. The …………… speed of an object is its speed at a particular instant of time.
5. The …………… of an object moving in a straight line is a measure of the rate at which it changes speed.
6. The average acceleration of an object can be calculated by …………… the change in its speed by the time taken for the change.
7. The force of gravitational attraction to a large object such as a planet is called …………… .
8. An object will remain at rest, or will not change its speed or direction, unless it is acted upon by an outside, …………… force.
9. The acceleration of an object depends on the mass of the object and the …………… force acting on it.
10. When an object applies a force to a second object, the second object applies an …………… and …………… force to the first object.
11. …………… is done whenever a force is applied to an object and the object moves in the direction of the force.
12. …………… can be transferred to an object by doing work on it. Doing work on an object can also convert the energy an object possesses from one form into another.
13. All stored energy is called …………… energy. …………… potential energy and …………… potential energy are examples of stored energy.
14. The Law of Conservation of Energy states that energy is never created or …………… .

average	destroyed	potential	weight
energy	equal	total	opposite
direction	dividing	work	gravitational
unbalanced	instantaneous	acceleration	
elastic	velocity	rate	

Answers

Speed and velocity

Worksheet 8.1

Answers

Calculating average speed

1. The average speed (v) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

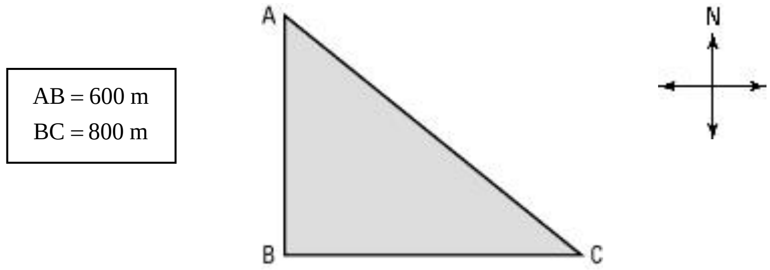
where d = distance travelled and t = time taken.

Complete the following table. Make sure you include the appropriate units for each quantity.

Distance travelled	Time taken	Average speed
100 m	20 s	$\frac{100}{20}$ = 5 m/s
3 km	$\frac{3}{60}$ = 0.05 h	60 km/h
20 × 15 = 300 m	15 s	20 m/s
2 km	$\frac{2000}{25}$ = 80 s	25 m/s

Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.



Both vehicles take 70 s for the trip.
Perform the necessary calculations to complete the table.

Average speed of X	Average velocity of X	Average speed of Y	Average velocity of Y
d = 800 + 600 = 1400 m t = 70 s $v = \frac{d}{t} = \frac{1400}{70}$ = 20 m/s	$v = \frac{\text{displacement}}{\text{time}} = \frac{1000}{70}$ = 14.3 m/s Direction = 37 °S of E (draw a scale diagram and measure the angle)	d(AC) = 1000 m (this is a 3:4:5 right-angled triangle) t = 70 s $v = \frac{d}{t} = \frac{1000}{70}$ = 14.3 m/s	$v = \frac{\text{displacement}}{\text{time}} = \frac{1000}{70}$ = 14.3 m/s Direction = 37 °S of E (draw a scale diagram and measure the angle)

Ticker tapes

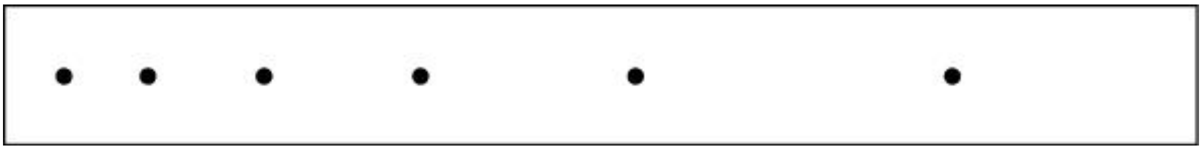
Worksheet 8.2

Answers

Describing motion

1. For each ticker tape shown, describe the motion of the body to which the tape was attached. The left-hand dot is at time zero.

(a)



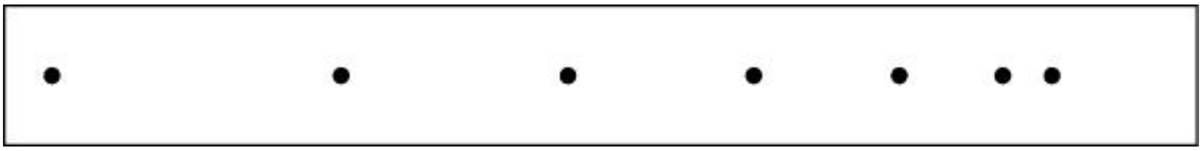
The speed of the body is increasing with time. It is accelerating.

(b)

Initially, the body accelerates (its speed is increasing) and then rapidly decelerates (its speed is decreasing).

Measuring speed

2. The time between each of the dots on the following ticker tape is 0.02 s.



(a) Calculate the average speed of the object to which the tape was attached for each time interval. Express your answers in mm/s.

This is a sample calculation; student answers may vary depending on the magnification of the image.

Interval	Distance between dots (mm)	Average speed (mm/s)
AB	38	$\frac{38}{0.02}$ = 1900 mm/s
BC	30	$\frac{30}{0.02}$ = 1500 mm/s
CD	25	$\frac{25}{0.02}$ = 1250 mm/s
DE	20	$\frac{20}{0.02}$ = 1000 mm/s
EF	14	$\frac{14}{0.02}$ = 700 mm/s
FG	7	$\frac{7}{0.02}$ = 350 mm/s

(b) Describe the motion of the object to which this tape was attached.

The average speed of the object is decreasing with time. The object is decelerating.

Worksheet 8.3

Acceleration

Answers

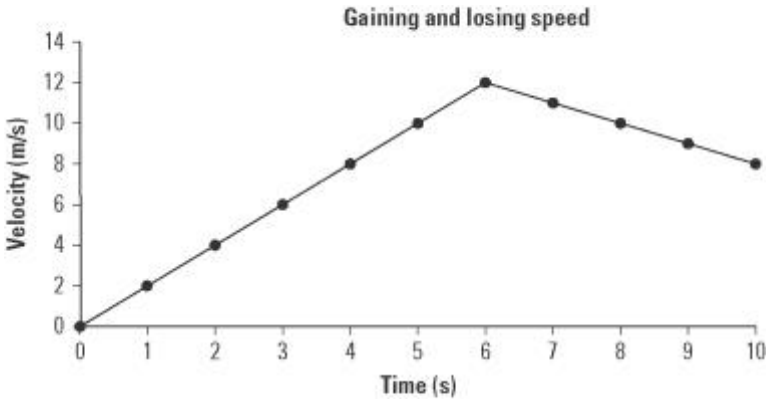
The velocities of two objects were recorded over ten-second time periods, as shown in the examples below.

Example: Gaining and losing speed

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	2	4	6	8	10	12	11	10	9	8

1. (a) Use a spreadsheet application such as Excel to plot a line graph of this data.

(b) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units. Use the graph to answer the following questions.



(c) Explain the shape of the graph.

The object accelerates uniformly for the first six seconds and then decelerates uniformly for the next four seconds.

(d) At the six-second mark the object's speed begins to decrease. If the object's speed continues to decline at the same rate after ten seconds, at what time will its velocity be zero?

At t = 18 seconds its velocity will be zero as its velocity is decreasing by 1 m/s every second.

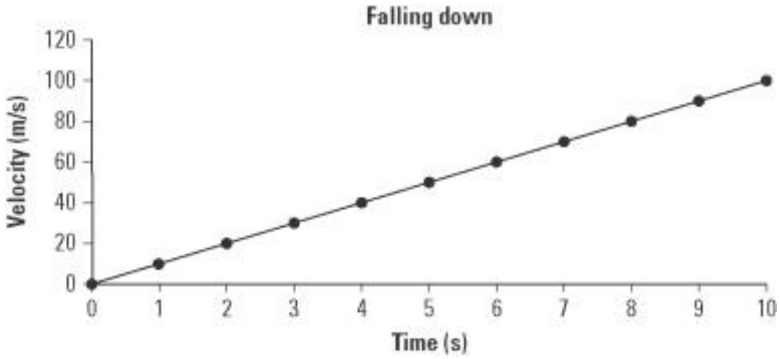
Example: Falling down

Time (s)	0	1	2	3	4	5	6	7	8	9	10
Velocity (m/s)	0	9.8	19.6	29.4	39.2	49.0	58.8	68.6	78.4	88.2	98.0

2. (a) Examine the data table and determine the pattern in the data. Use this pattern to complete the table.

(b) Use a spreadsheet application such as Excel to plot a line graph of this data.

(c) Format your graph. Ensure that the graph has a title and that each axis has a title and appropriate units. Use the graph to answer the following questions.



(d) Explain how this graph shows that falling due to gravity is an example of uniform acceleration.

The graph is linear with a constant slope. Thus, the falling object's velocity is increasing at a uniform rate as would be expected for a body falling in a gravity field (as long as friction is absent or negligible).

(e) The slope of the graph is equal to the acceleration due to gravity. Determine the slope of the graph in m/s^2.

$$\text{Slope} = \frac{\text{rise}}{\text{run}} = \frac{98}{10} = 9.8\ \text{m/s}^2$$

Inertia and motion

Answers

1. State Newton's First Law of Motion.

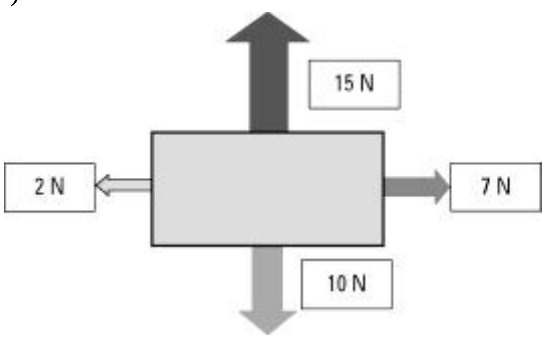
A body at rest will remain at rest and a body in motion will not change its speed or direction unless acted upon by an unbalanced force.

2. Consider the following examples of forces acting on a body. In each case find the net (unbalanced) force and describe the resulting motion of the object. Assume north is up the page.

(a)

The body will move to the left as the net force is 20 – 10 = 10 N to the left (i.e. west).

(b)

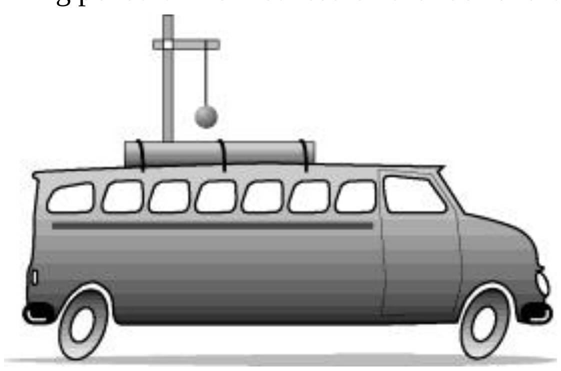


The body will move at 45° to the right of north (i.e. north-east). The net horizontal force is 7 – 2 = 5 N to the right (east). The net vertical force is 15 – 10 = 5 N up the page (north). The net result of these two equal forces is motion towards the north-east.

3. A 500-kg box is placed on the tray of a utility truck but it is not secured by ropes. The ute drives along a highway, then the driver has to brake suddenly to avoid a wombat on the road. Use the concept of inertia to explain the danger to the driver.

Inertia is a measure of a body's resistance to a change in motion. The unsecured box is travelling with the same speed as the ute. When the ute suddenly brakes, the inertia of the heavy box will result in it continuing to move forward into the cabin of the ute. This could cause severe injury or death to the driver.

4. A 1-kg pendulum is mounted on the roof of a vehicle as shown in the diagram.



Describe the motion of the pendulum in each example.

(a) The vehicle accelerates from rest.

The bob will hang backwards towards the rear of the vehicle.

(b) The vehicle brakes from a speed of 60 km/h.

The bob will hang forward towards the front of the vehicle.

Worksheet 8.5

Force and gravity

Answers

Weight

1. The weight of an object is the force acting on the object in a gravitational field. The stronger the gravitational field, the greater the weight of a body. Weight is measured in the units of Newtons (N)

The weight (W) of a body of mass (m) in a gravitational field (g = acceleration due to gravity) is given by the formula:

$$W = mg$$

On the Earth's surface 'g' typically has a value of 9.8 m/s^2 but this varies with altitude and latitude. On other planets, the value of 'g' also changes.

Complete the table using the above formula.

Mass (kg)	Gravity (m/s^2)	Weight (N)
20	9.8	$20 \times 9.8 = 196$
30	$\frac{286.5}{30} = 9.55$	286.5
$\frac{400}{9.8} = 40.8$	9.8	400.0

Spreadsheets and graphing

2. A spreadsheet program such as Excel is a useful computer application for tabulating data and plotting graphs. In this exercise you will practise using a spreadsheet application.

The following mass and weight data was collected from the planet Mars. The data is presented as you would set it out in the spreadsheet application.

(a) Open the spreadsheet application and enter the data as tabulated below.

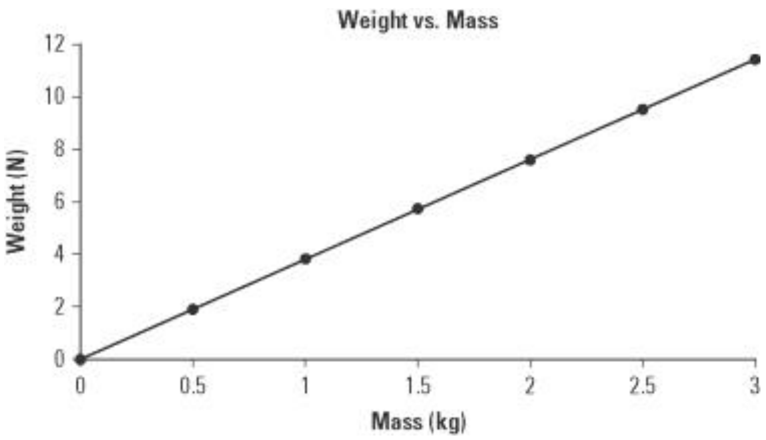
(b) Select *Insert* from the menu bar.

(c) Select *Scatter* as the chart type. Select 'Scatter with straight lines and markers'.

(d) The *Design* and *Layout* tabs can be used to insert a chart title and then each axis title and unit.

(e) Select the gridlines you would like to show.

	A	B
1	Mass (kg)	Weight (N)
2	0.0	0.0
3	0.5	1.9
4	1.0	3.8
5	1.5	5.7
6	2.0	7.6
7	2.5	9.5
8	3.0	11.4



Use your chart to answer the following questions.

(f) Estimate the weight of a 1.3 kg body on the surface of Mars. ....4.9 N.................

(g) Estimate the mass of an object that weighs 10.0 N on Mars. ....2.6 kg.............

(h) Suggest what information the slope of the graph may provide.

(*Clue:* The formula $W = mg$ can be rearranged as $g = \frac{W}{m}$.)

The slope provides the acceleration due to gravity on the planet. ....

....

Worksheet 8.6

Science Quest 10: pages 316–17

Newton's Second Law

Answers

1. State Newton's Second Law of Motion.

The acceleration of a body is directly proportional to the net force and inversely proportional to its mass.

2. Use the formula: $F = ma$ to complete the following table

Force (F) (newtons)	Mass (m)	Acceleration (a)
100	40 kg	$\frac{100}{40} = 2.5\ \text{m/s}^2$
2	200 g	$\frac{2}{0.20} = 10\ \text{m/s}^2$
70	$\frac{70}{3.5} = 20\ \text{kg}$	$3.5\ \text{m/s}^2$
100	$\frac{100}{0.10} = 1000\ \text{kg}$	$10\ \text{cm/s}^2$

3. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.

(a) Calculate the net force acting on the body.

$F = 35 - 15 = 20\ \text{N}$

(b) Calculate the acceleration of the body.

$a = \frac{F}{m} = \frac{20}{40} = 0.5\ \text{m/s}^2$

4. The photo shows a drag car coming to a stop at the end of a race.

Photo: Claudia Schnepf / 123RF

(a) To come to a complete stop, the drag car negatively accelerates. Identify another term for 'negative acceleration'. Deceleration

(b) Identify the force that brings the drag car to a stop. Friction

Worksheet 8.7

Science Quest 10: pages 318–19

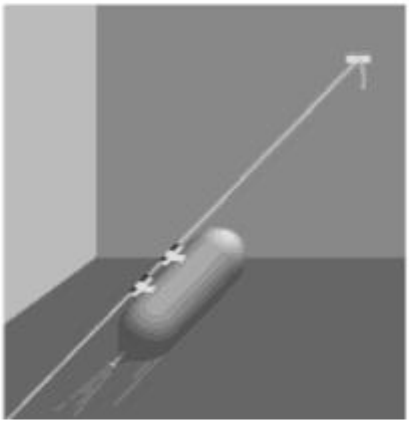
Newton's Third Law

Answers

1. State Newton's Third Law of Motion.

All forces occur in pairs; the forces are equal in magnitude but opposite in direction.

2. The following diagram shows a balloon rocket built by students to investigate motion. The balloon is mounted on a horizontal string using a length of drinking straw and masking tape.



The inflated balloon is then released.

(a) Where along the string will the balloon be moving at the greatest speed?

The greatest speed will occur when the balloon is initially released. This is when the force of the gas escaping is at its greatest and the reaction force acting on the balloon is at its greatest.

(b) Use Newton's Third Law to explain why the balloon is propelled forward along the string.

The contracting rubber balloon pushes the air out of the open end of the balloon. The force of gases escaping from the balloon leads to a reaction force in which the gases push back on the balloon, which pushes it forward.

3. The photograph below shows a person pushing against a wall.

Photo: Wavebreak Media Ltd / 123RF

The man is not moving even though he is pushing forward against the wall.

(a) What is the net force acting on the man?

Zero (he is not moving)

(b) If the man pushes with a force of 200 N on the wall, what is the force that the wall exerts back on him?

200 N (but in the opposite direction)

(c) The man is pushing back on the floor with his feet with a force 200 N. With what force does the floor push back on the man?

200 N (but in the opposite direction)

(d) Explain what would happen if he tried to do this on a very slippery floor.

As he pushes back against the slippery floor the lack of friction causes his foot to slide as the reaction force of the floor back on the shoe is too low. The low value of this force and the force of the wall pushing back on the man will cause him to fall down.

Worksheet 8.8

Forces, energy and motion: Summary

Answers

Complete the statements below to compile a summary of this unit. The missing words can be found in the word list below.

1. Speed is a measure of the …rate… at which an object moves over a distance.
2. …Average… speed can be calculated by dividing the distance travelled by the time taken.
3. …Velocity… is a measure of the rate of change in position. In order to describe the velocity fully, the …direction… of the change in position must be stated.
4. The …instantaneous… speed of an object is its speed at a particular instant of time.
5. The …acceleration… of an object moving in a straight line is a measure of the rate at which it changes speed.
6. The average acceleration of an object can be calculated by …dividing… the change in its speed by the time taken for the change.
7. The force of gravitational attraction to a large object like a planet is called …weight… .
8. An object will remain at rest, or will not change its speed or direction, unless it is acted upon by an outside, …unbalanced… force.
9. The acceleration of an object depends on the mass of the object and the …total… force acting on it.
10. When an object applies a force to a second object, the second object applies an …equal… and …opposite… force to the first object.
11. …Work… is done whenever a force is applied to an object and the object moves in the direction of the force.
12. …Energy… can be transferred to an object by doing work on it. Doing work on an object can also convert the energy an object possesses from one form into another.
13. All stored energy is called …potential… energy. …Gravitational… potential energy and …elastic… potential energy are examples of stored energy.
14. The Law of Conservation of Energy states that energy is never created or …destroyed… .

average	destroyed	potential	weight
energy	equal	total	opposite
direction	dividing	work	gravitational
unbalanced	instantaneous	acceleration	
elastic	velocity	rate	